BAIT3153 Software and Project Management (SPM)

Chapter 6

# Software Process Improvement

# Software Process Improvement

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- Why Process Improvement
- Quality Factors (**what are the factors that affect product quality?**)
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## Why Process Improvement?

**Input □ Process □ Output** *(<u>software & documentations</u>)*

- **A good process** is usually required to **produce a good product**, and process improvement benefits arise because the quality of the product depends on its development process.
- **Software companies carry out software process improvement with the hopes of achieving the following benefits:**

✧To <u>increase</u> software **quality**

✧To <u>reduce</u> development **time**

✧To <u>reduce</u> development **costs**


# Quality Factors: What <u>factors</u> determine product quality?

| | |
|---|---|
| <u>**Large**</u> **Projects** | Input □ **Process** □ Output<br>*(Development **Process**)* *(software)*<br>*(*developers capabilities average)* |
| <u>**Small**</u> **Projects** | **Input** □ Process □ Output<br>*(Capabilities of **Developers**)* *(software)*<br>*(Development **Technology**)* |

***If schedule is unrealistic*, both large & small project's QUALITY will suffer**

What <u>factors</u> determine product quality? ____


# Software Process Improvement Considerations

## 6.1 The Rationale for Process Improvement

- There is no one-size-fits-all or ‘ideal’ or ‘standard’ software development **process**

- Every company has to create its own **process** depending on:
  - the company’s size
  - staff’s background and skills
  - type of software being developed
  - customer and market requirements
  - local environment
  - company culture, etc


# Software Process Improvement Considerations

## 6.1 The Rationale for Process Improvement

- When a piece of work is completed, copies are distributed to co-workers who examine the work, noting defects/problems. A meeting then discusses the work and a list of defects requiring rework is produced.
- E.g. of works that may be examine:
  - Table design (e.g. fields in a table – too little?, too many?)
  - Program codes (e.g. length of variable names, etc)
  - \-\-\-\-\-\-\-\-\-\-\-\-\-\-

- **For process improven**
  - **If the goal is to imp** consider:
    - introducing **new** activities (e.g. see above)
    - changing the **way** software is developed and tested
  - **If your goal is to improve the "process" itself, select the most important process attributes.**


# Software Process Improvement Considerations

## 6.1 The Rationale for Process Improvement

### Process Attribute

| Process attribute | Key issues |
|---|---|
| Understandability | To what extent is the process explicitly defined and how easy it is to understand the process definition? |
| Standardization | To what extent is the process based on a standard generic process? This may be important for some customers who require conformance with a set of defined process standards. To what extent is the same process used in all parts of a company? |
| Visibility | Do the process activities culminate in clear results, so that the progress of the process is externally visible? |
| Measurability | Does the process include data collection or other activities that allow process or product characteristics to be measured? |
| Supportability | To what extent can software tools be used to support the process activities? |


# Software Process Improvement Considerations

6.1 The Rationale for Process Improvement

## Process Attribute

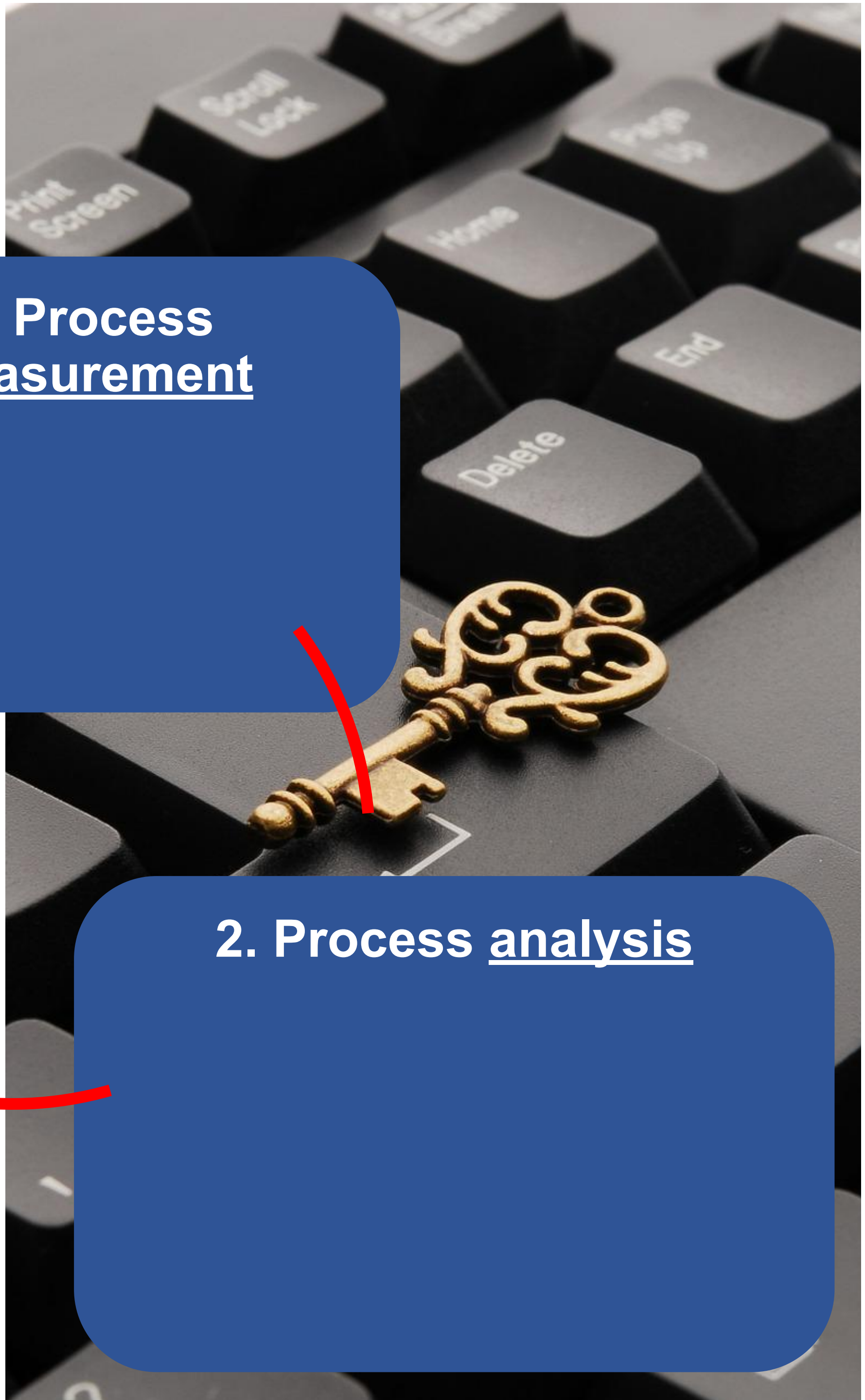
| Process attribute | Key issues |
|---|---|
| Acceptability | Is the defined process acceptable to and usable by the engineers responsible for producing the software product? |
| Reliability | Is the process able to avoid or trap errors before they result in product defects? |
| Robustness | Can the process continue in spite of unexpected problems? |
| Maintainability | Can the process evolve to reflect changing organizational requirements or identified process improvements? |
| Rapidity | How fast can the process of delivering a system from a given specification be completed? |

## 6.2 Software Process Improvement Cycle

1. Process measurement

2. Process analysis

3. Process change



## 6.2 Software Process Improvement Cycle

**1. Process measurement**

**Measure** relevant attributes of current process. These measurement will be the baseline for assessing

**measure the same attribute again!**

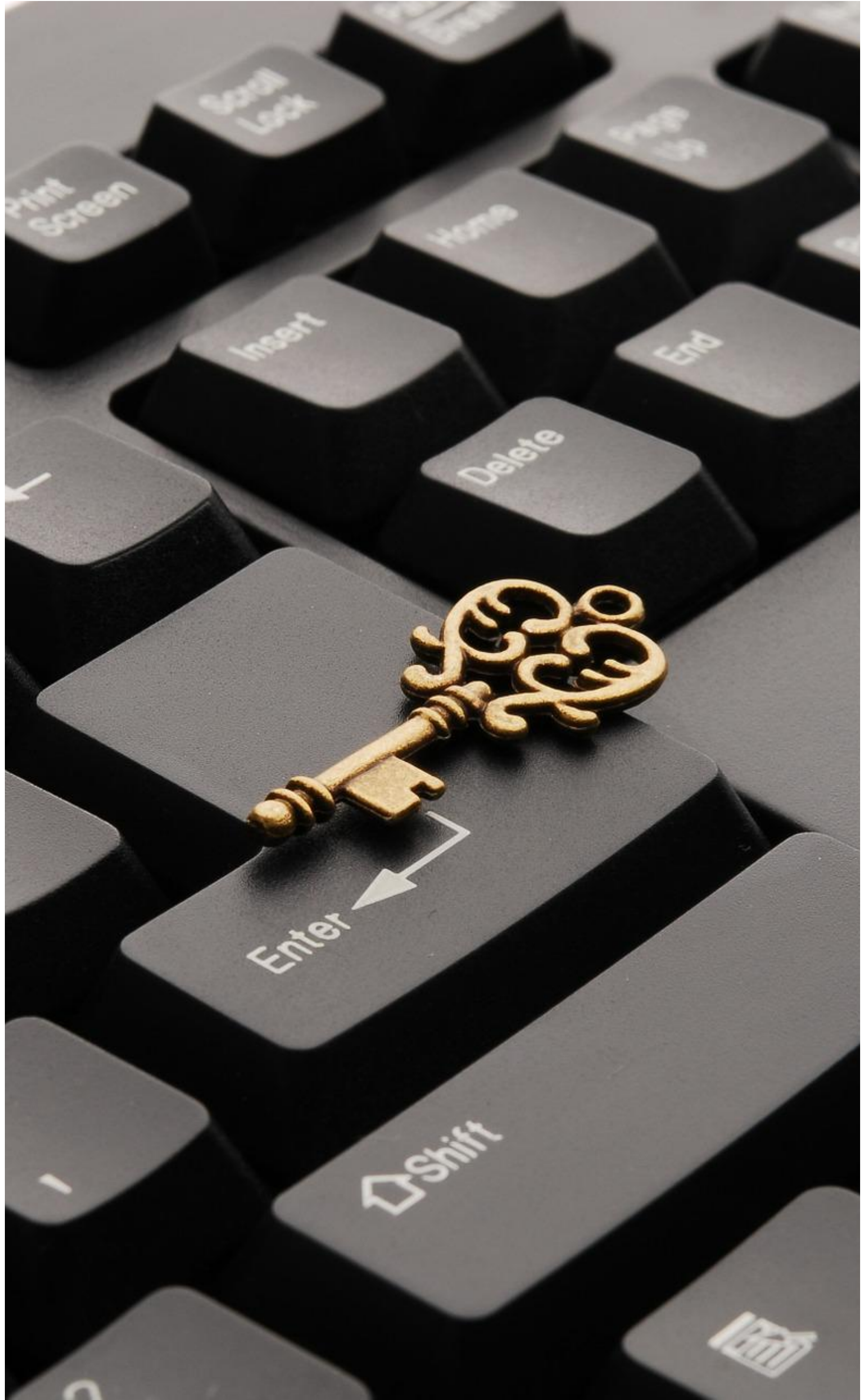
**2. Process analysis**

Analyse current process to identify bottlenecks and weaknesses.

(*note: Process models may be develop during this stage*

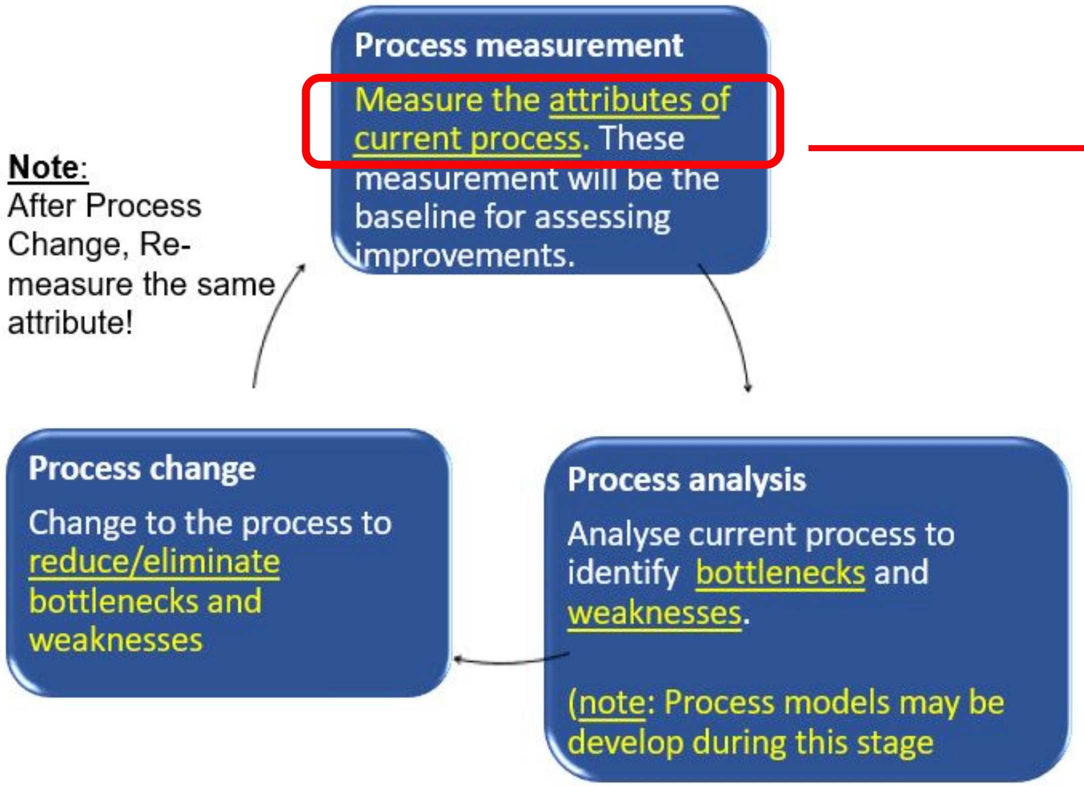
**3. Process change**

Change the process to reduce/eliminate bottlenecks and weaknesses



# Process Measurement

## 6.2 Software Process Improvement Cycle




The types of <u>process metrics</u> that can be collected during process measurement:

1. <u>**Time**</u> **taken for process activities to be completed**

   ***Examples:***

   - *<u>time</u> to complete "requirements gathering",*
   - *<u>time</u> to complete UI design,*
   - *<u>time</u> to complete the software dev. process from start to finish,*

2. <u>**Resources**</u> **required for processes or activities**

   **(e.g. people, hardware and software)**

3. **Number of occurrences of a particular <u>event</u>**

   ***e.g.1.*** ***no of defects*** *discovered 1 week after delivery*

   ***e.g.2.*** ***no of run-time errors*** *encountered during the 1st UAT*

   ***e.g.3.*** ____

4. *Others (e.g. ____ )* **Understandability, Visibility, Acceptability, Reliability, …**

# Process Measurement

## 6.2 Software Process Improvement Cycle

**Process measurement**

Measure the attributes of current process. These measurement will be the baseline for assessing improvements.

**Note**:
After Process Change, Re-measure the same attribute!

**Process change**

Change to the proc
reduce/eliminate
bottlenecks and
weaknesses

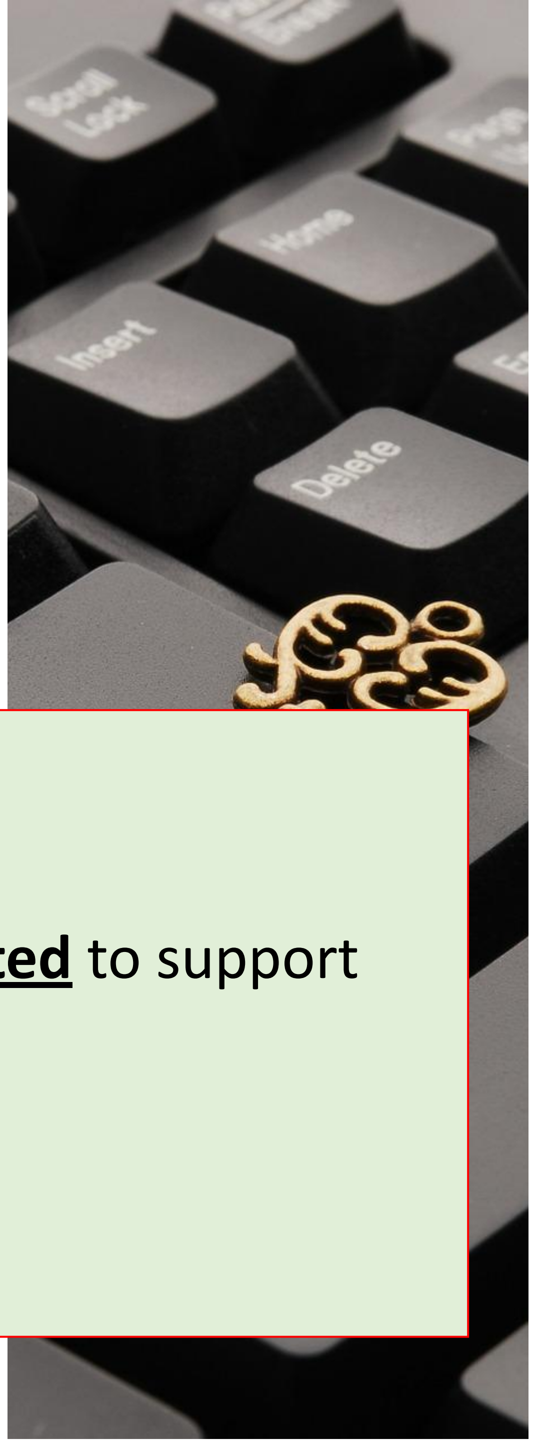
Collect process metrics

- ***Time*** taken for process activities to be completed
- **Resources** required for processes or activities
- Number of occurrences of a particular **event**
- **Others (e.g.** Understandability, Visibility, Acceptability, Reliability, … **)**

**Difficulty in process measurement:**

What information about the process should be **collected** to support process improvement?

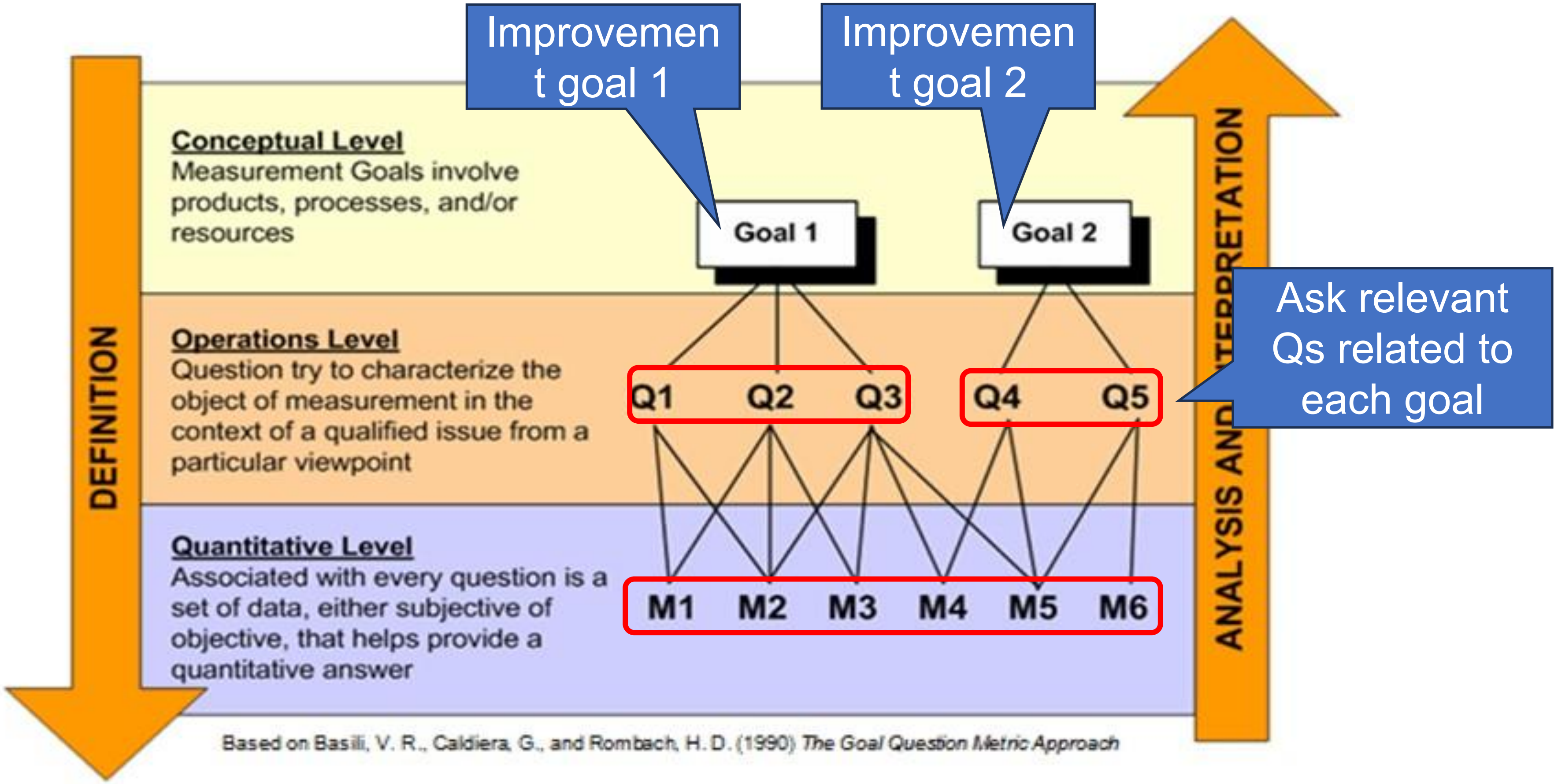
**Use GQM** to solve this difficulty.
*[GQM = **Goal-Question-Metric (GQM)** ]*

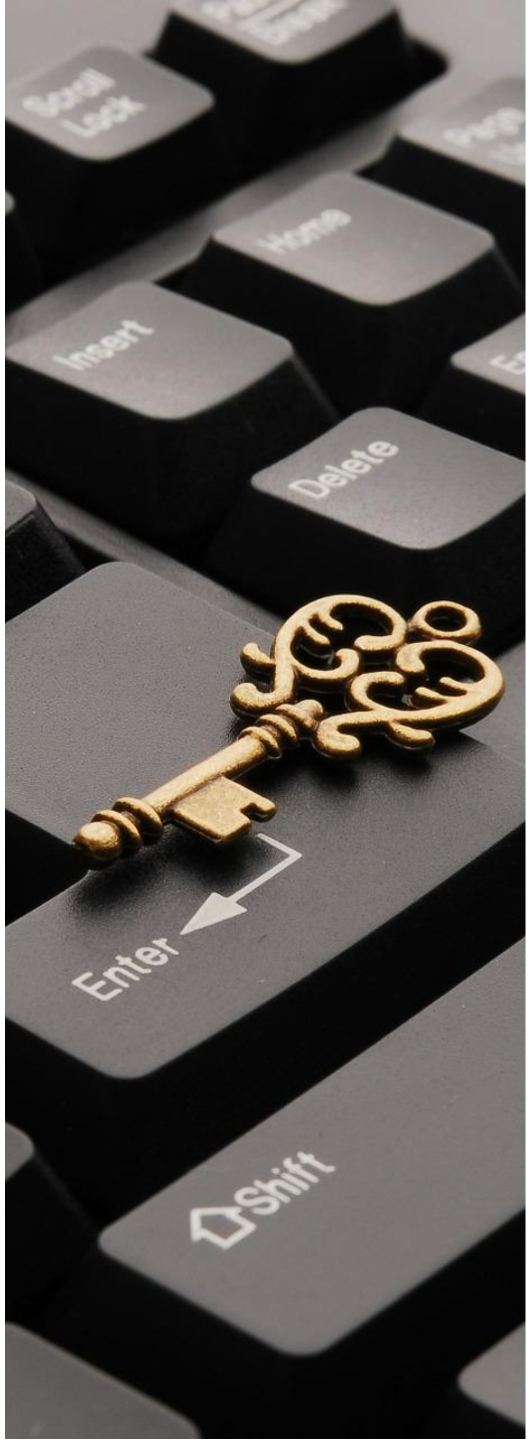


# Process Measurement

6.2 Software Process Improvement Cycle

## Goal-Question-Metric (GQM) Paradigm



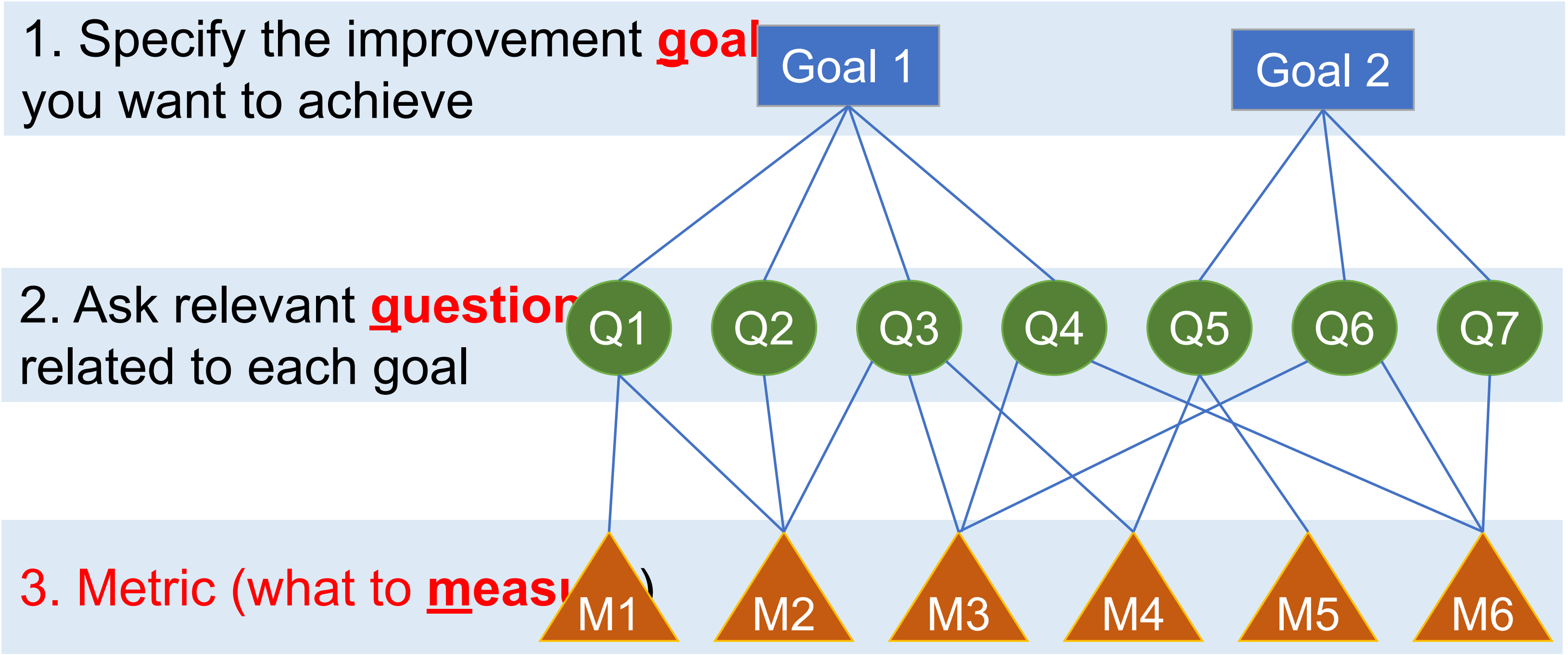



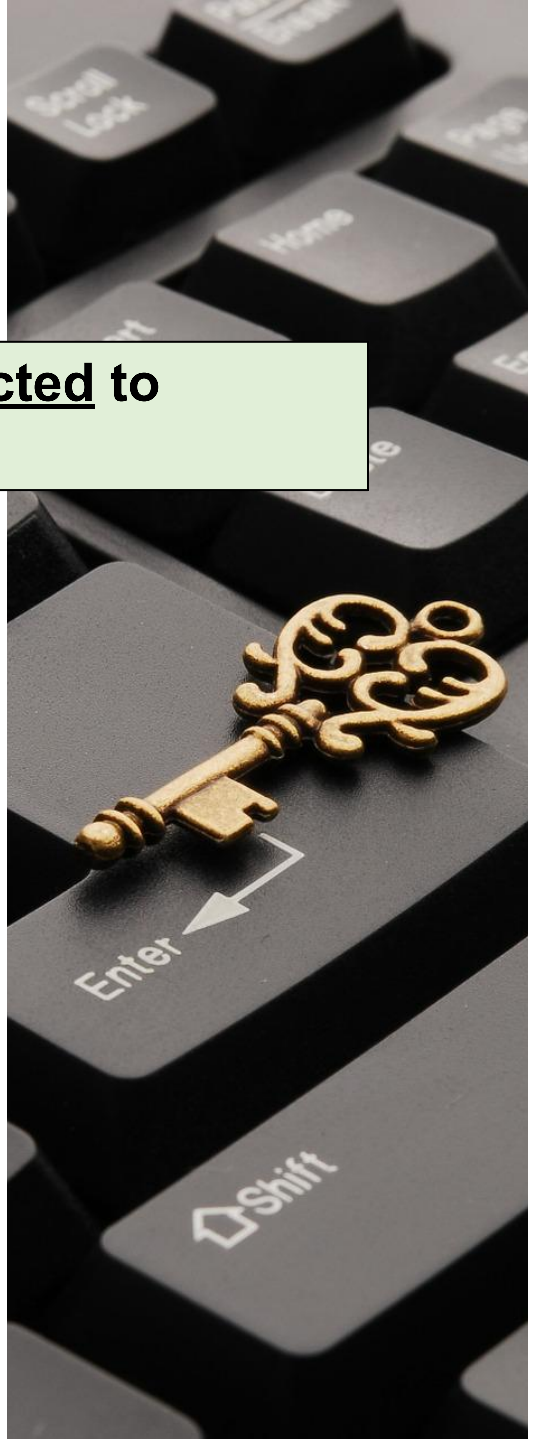
# Process Measurement

6.2 Software Process Improvement Cycle

## **Goal-Question-Metric (GQM)** Paradigm

**Use GQM to determine what information about the process should be collected to support process improvement**



# Process Measurement

6.2 Software Process Improvement Cycle

## **Goal-Question-Metric (GQM)** Paradigm

**Use GQM to determine <u>what</u> information about the process should be collected to support process improvement**

## Example 1

1. Specify the **improvement g**
You want to achieve

**<u>Reduce efforts in software maintenance</u>**

2. Ask **relevant <u>questions</u>** related to the goal

**How big is the software?**

**How easy it is to understand the software?**

3. Metric (what to **<u>meas</u>**

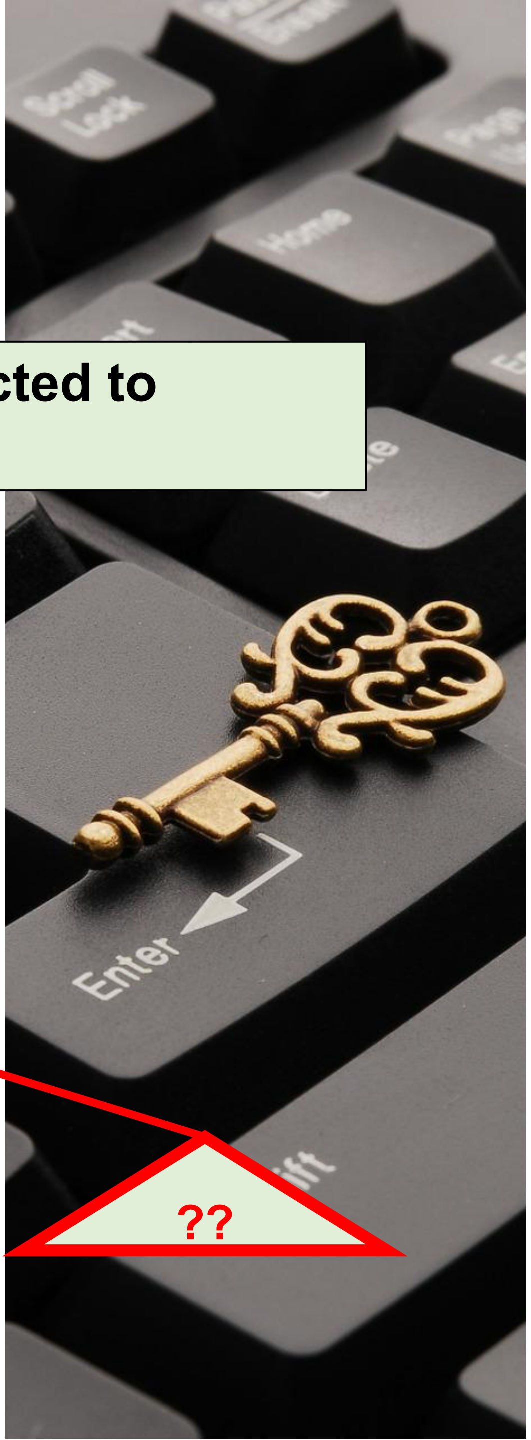
**LOC**

**No of methods**

**No of comment lines**

**Length of var names**

**??**



# Process Measurement

6.2 Software Process Improvement Cycle

## **Goal-Question-Metric (GQM)** Paradigm

**Use GQM to determine <u>what</u> information about the process should be collected to support process improvement**

### **Example 2**

1. Specify the **improvement goals** You wants to achieve

**<u>Improve the performance</u>** of an e-commerce website

2. Ask **relevant <u>questions</u>** related to the goal

What is the current page load time?

How many users abandon their carts before completing a purchase?

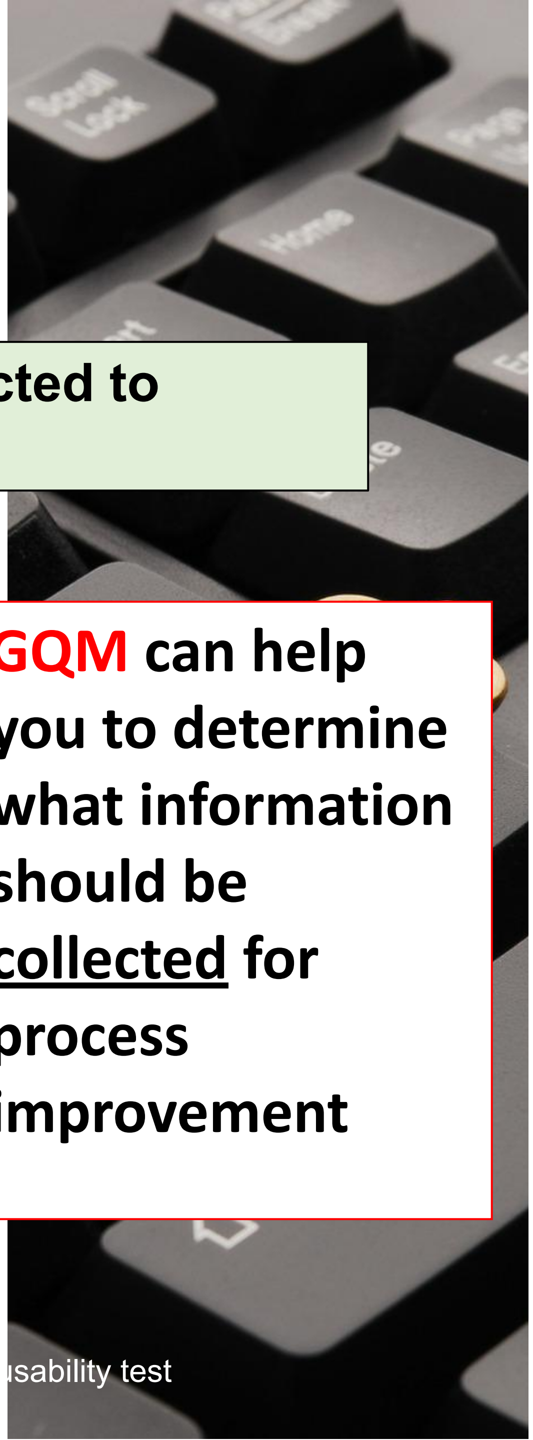
3. Metric (what to **<u>measure</u>**)

Page load time (e.g. 5s)

cart abandonment rate (e.g. 20/day)

Q) Suggest how to reduce page load time? Use smaller pic size

Q) Suggest how to reduce cart abandonment? Consider carrying out usability test before system roll-out



**GQM can help you to determine what information should be <u>collected</u> for process improvement**

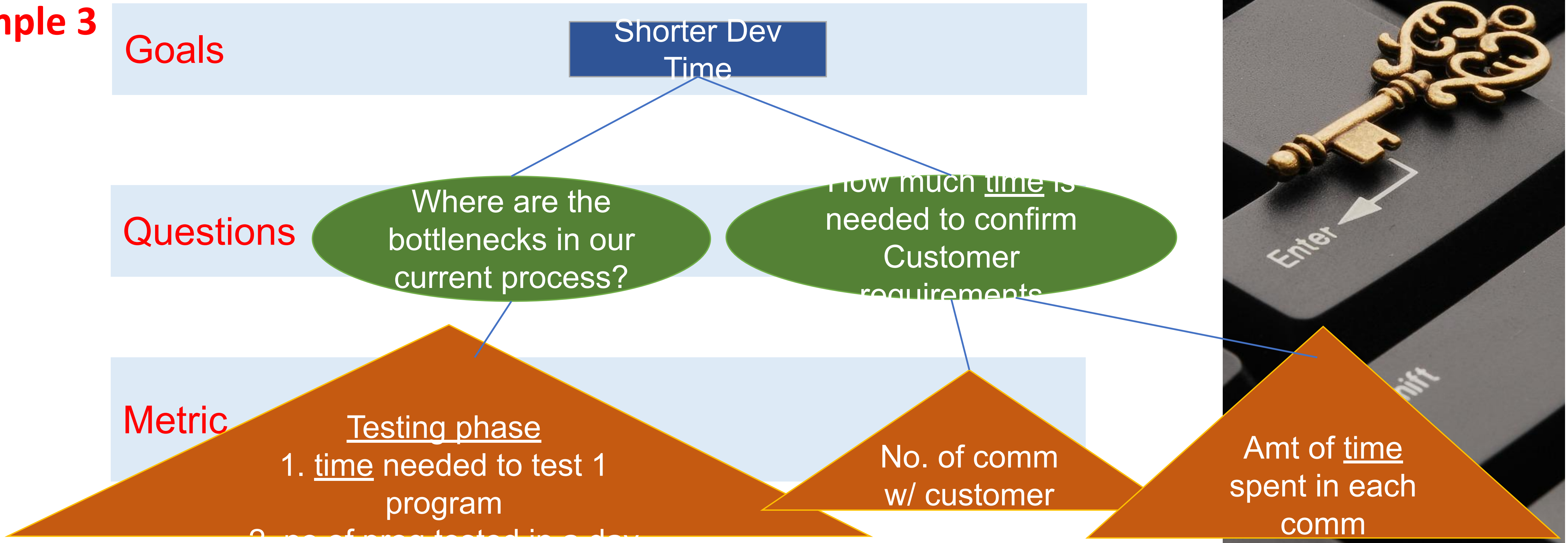
# Process Measurement

6.2 Software Process Improvement Cycle

## Goal-Question-Metric (GQM) Paradigm

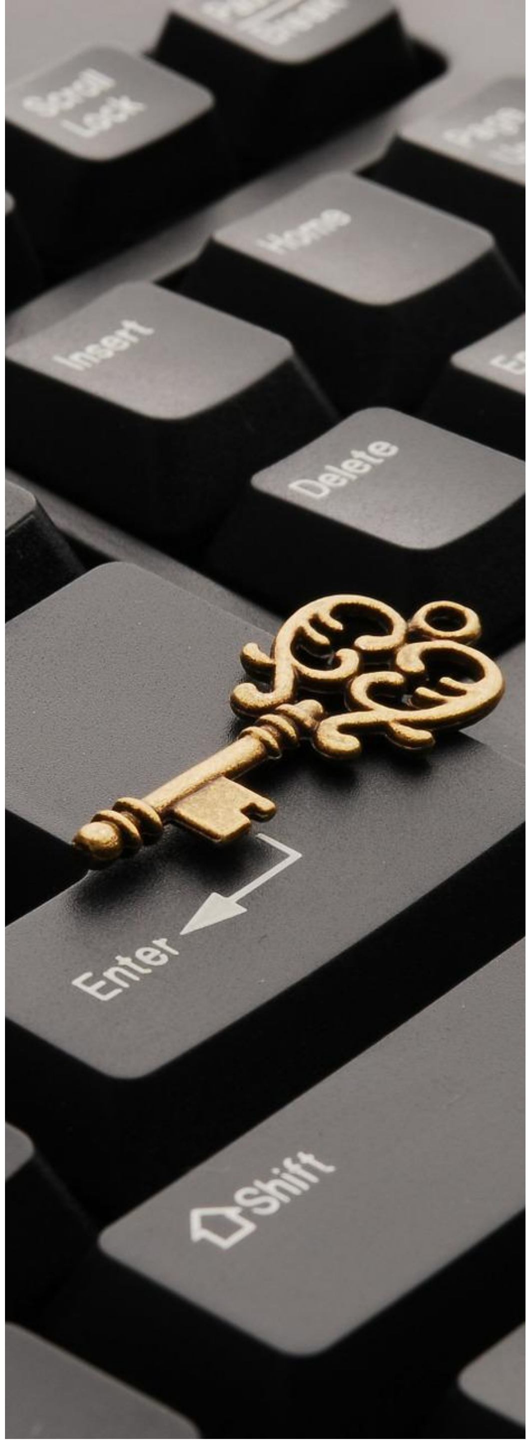
**Use GQM to determine <u>what</u> information about the process should be collected to support process improvement**

**Example 3**



## Goal-Question-Metric (GQM) Paradigm – Components

- **Goals**
  - What is the organisation trying to achieve? The objective of process improvement is to satisfy these goals.
  - Focus on how the process affects products or the organization itself.
  - Examples:
    - **shorter product development time**
    - **increased product reliability.**
    - **improve process maturity level**



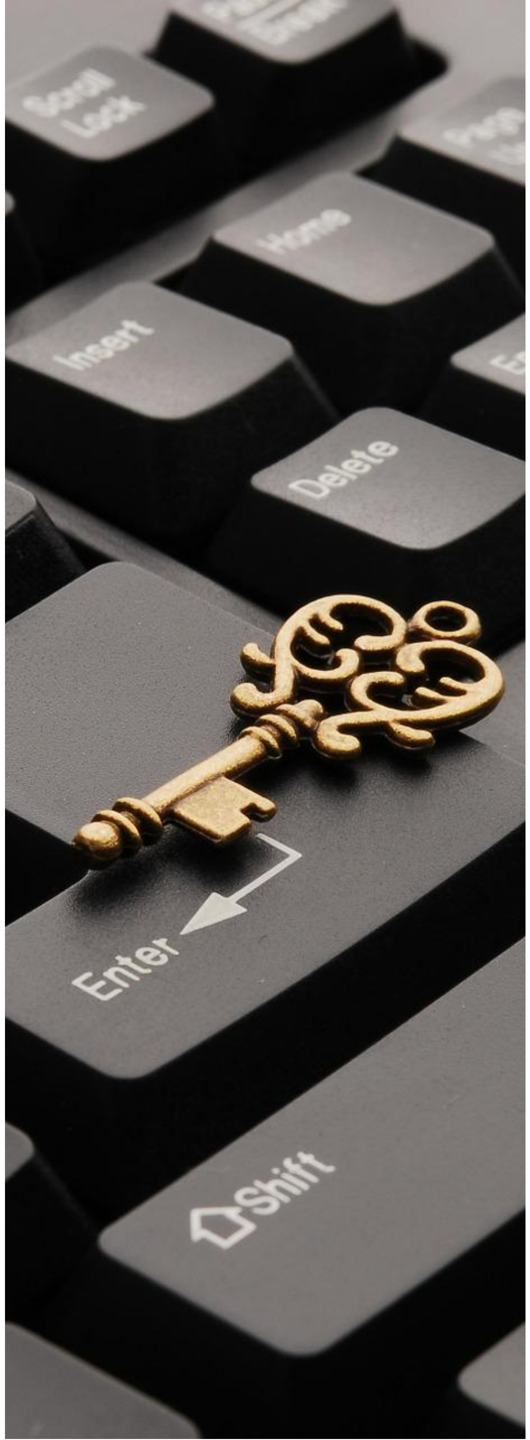
# Process Measurement

6.2 Software Process Improvement Cycle

## Goal-Question-Metric (GQM) Paradigm – Components

- **Goals**
  - What is the organisation trying to achieve? The objective of process improvement is to satisfy these goals.
  - Focus on how the process affects products or the organization itself.
  - Examples of goals:
    - shorter product development time
    - increased product reliability.
    - improve process maturity level

- **Questions**
  - Questions about areas of uncertainty related to the goals.
  - Examples of Questions (for the goal "*shorter development times*")
    - How much time is needed to finalize customer requirements?
    - How many errors are detected during UAT?



# Process Measurement

## 6.2 Software Process Improvement Cycle

### Goal-Question-Metric (GQM) Paradigm – Components

- **Goals**
  - What is the organisation trying to achieve? The objective of process improvement is to satisfy these goals.
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  - Examples of goals:
    - shorter product development time
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- **Questions**
  - Questions about areas of uncertainty related to the goals.
  - Examples of Questions (for the goal "*shorter development times*")
    - Where are the bottlenecks in our current process?
    - How much time is needed to finalize requirements with customers?

- **Metrics**
  - Collected to answer the questions and to confirm whether or not process improvements have achieved the desired goal.
  - Examples of measurements to be taken
    - time taken to complete each process activity
    - number of formal communications between clients and project manager to confirm requirements
    - number of errors detected during UAT

# Process Measurement

## 6.2 Software Process Improvement Cycle


### Goal-Question-Metric (GQM) Paradigm – Components

- **Goals**
  - What is the organisation trying to achieve? The objective of process improvement is to satisfy these goals.
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- **Metrics**
  - Collected to answer the questions and to confirm whether or not process improvements have achieved the desired goal.
  - Examples of measurements to be taken
    - time taken to complete each process activity
    - number of formal communications between clients and project manager for each requirements change
    - number of defects discovered per test run.

### GQM Paradigm – Advantages

- Separates organizational concerns (goals) from specific process concerns (questions)
- Provides a basis for deciding **what data should be collected for …**
- Suggests collect data should be analyzed in different ways, depending on the question it is intended to answer.

# Process Analysis

## 6.2 Software Process Improvement Cycle

Use GQM to decide what to measure

**Process measurement**

Measure the attributes of current process. These measurement will be the baseline for assessing improvements.

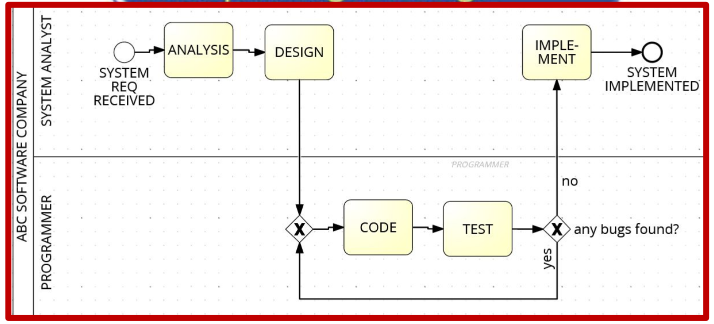
**Process analysis**

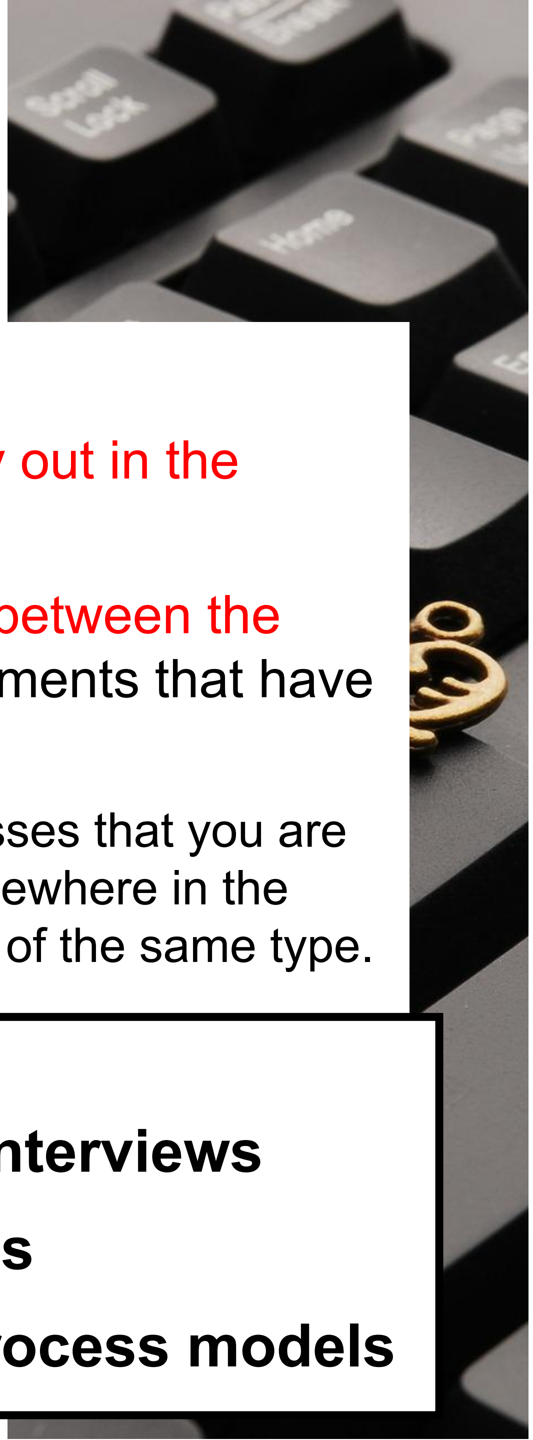
Analyse current process to identify bottlenecks and weaknesses.

(note: Process models may be develop during this stage

**Process change**

Change ~~to~~ the process to reduce/eliminate bottlenecks and weaknesses



## Objectives of Process Analysis

- **To understand** the **activities** carry out in the current process
- **To understand** the **relationships** between the process activities and the measurements that have been made.
- To relate the specific process or processes that you are analyzing to comparable processes elsewhere in the organization, or to idealized processes of the same type.

## Techniques

a. **Questionnaires and interviews**

b. **Ethnographic analysis**

c. **Analyse published process models**

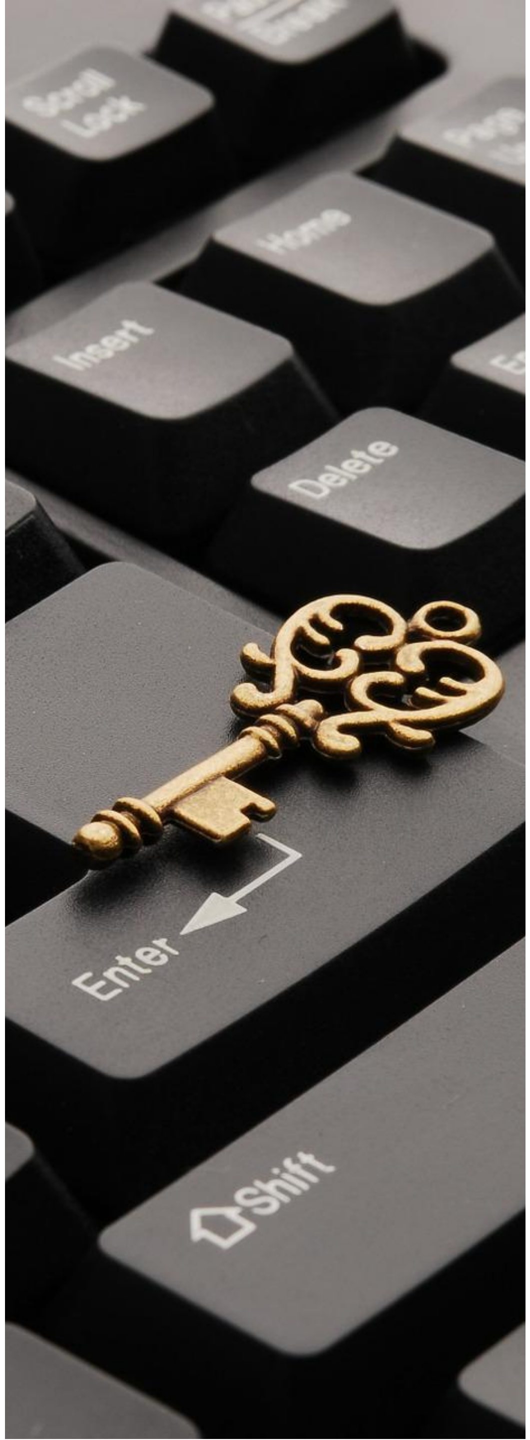
# Process Analysis - Techniques

6.2 Software Process Improvement Cycle

## a. Questionnaires and interviews

- **Questionnaires:**
  - ✔ Advantage: Fast
  - ✔ Disadvantage:
    - if questions are inappropriate □ incomplete/inaccurate understanding of the process
    - Participants (e.g. engineers and PM) treated it as assessment and give answer that they think you want to hear.
- **Interviews:**
  - ✔ Advantage: open-ended □ more information
  - x Disadvantage: ________

- Answers to a formal questionnaires are refined during personal interviews.

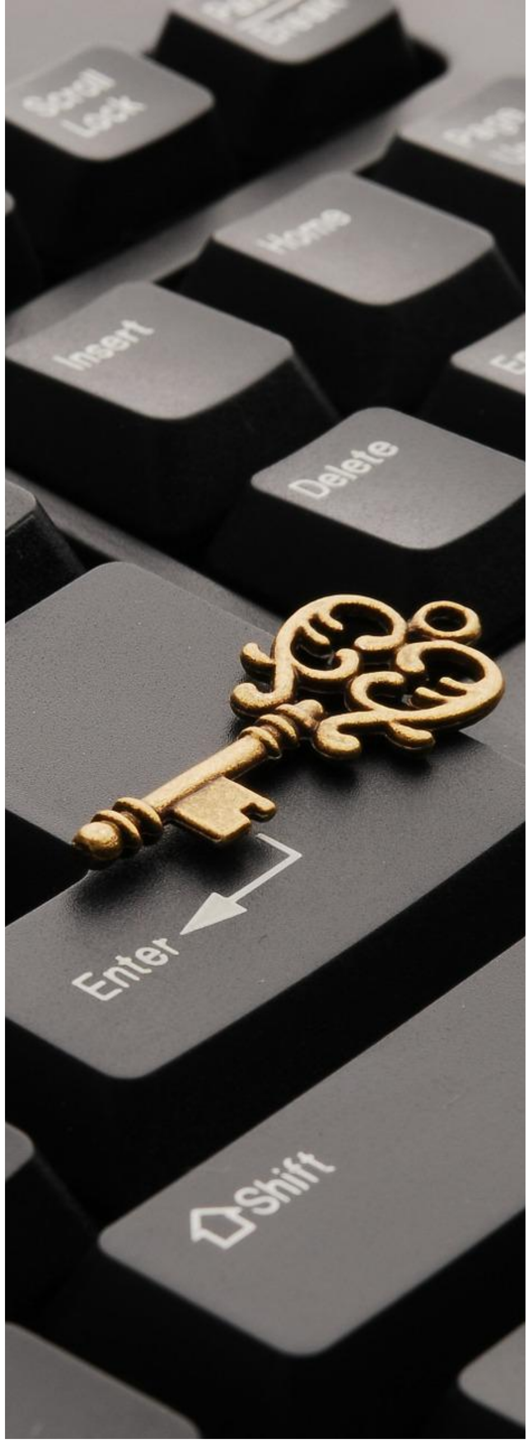


# Process Analysis - Techniques

6.2 Software Process Improvement Cycle

## b. Ethnographic analysis

- Involves assimilating process knowledge by <u>observation</u>.
- Best for in-depth analysis of process fragments rather than for whole-process understanding.
- The process fragments are chosen from the interview
- Disadvantage:
  - pro-longed process (several months)  from initial stages to maintenance.
  - Impractical for large projects (several years!!??)



# Process Analysis - Techniques

6.2 Software Process Improvement Cy

## Techniques

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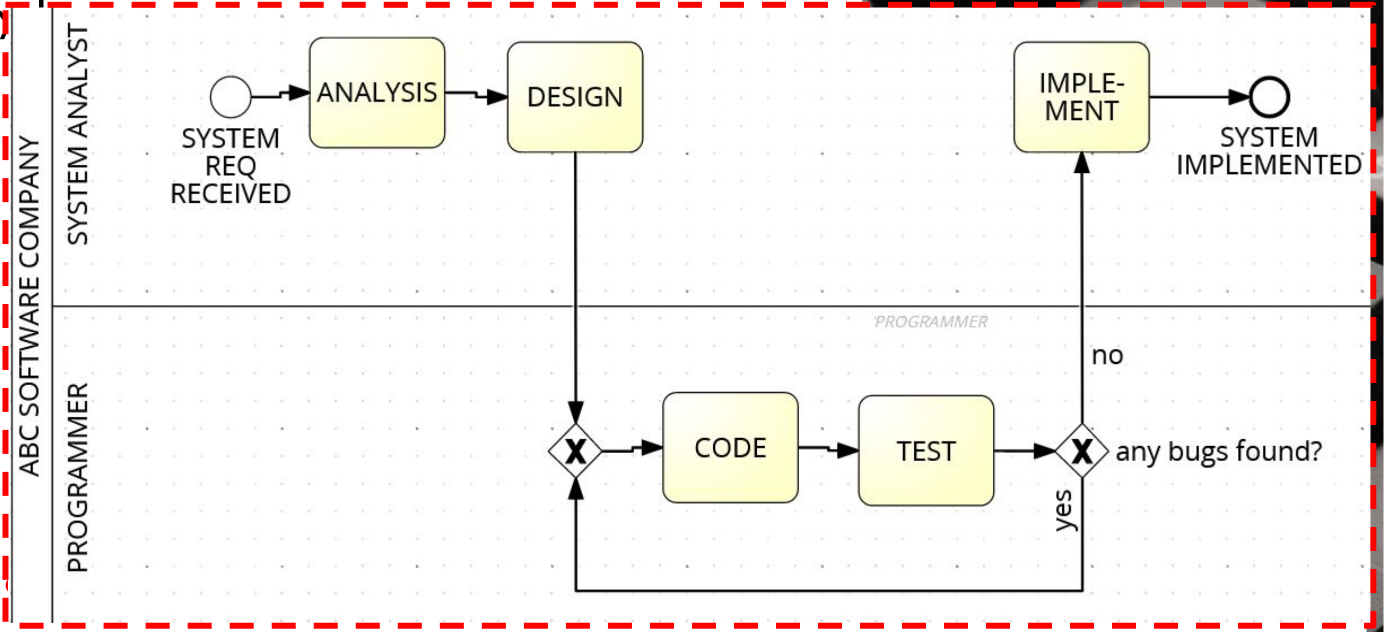



## c. Analyse Published process model

- Process models are a good way of focusing attention on the activities in a process and the information transfer between these activities.
- Process models do not have to be formal or complete –purpose is to provoke discussion rather than document the process in detail.
- Model-oriented **questions** can be used to help understand the process e.g.

1. Are there activities take place in practice but are not shown in the model?
2. Are there process activities, shown in the model, that you think are inefficient?
3. If something goes wrong, do you follow the model or take up emergency actions? Why or why not?
4. What are the methods of communication?
5. What tools are used to support the activities in the model? Effective?

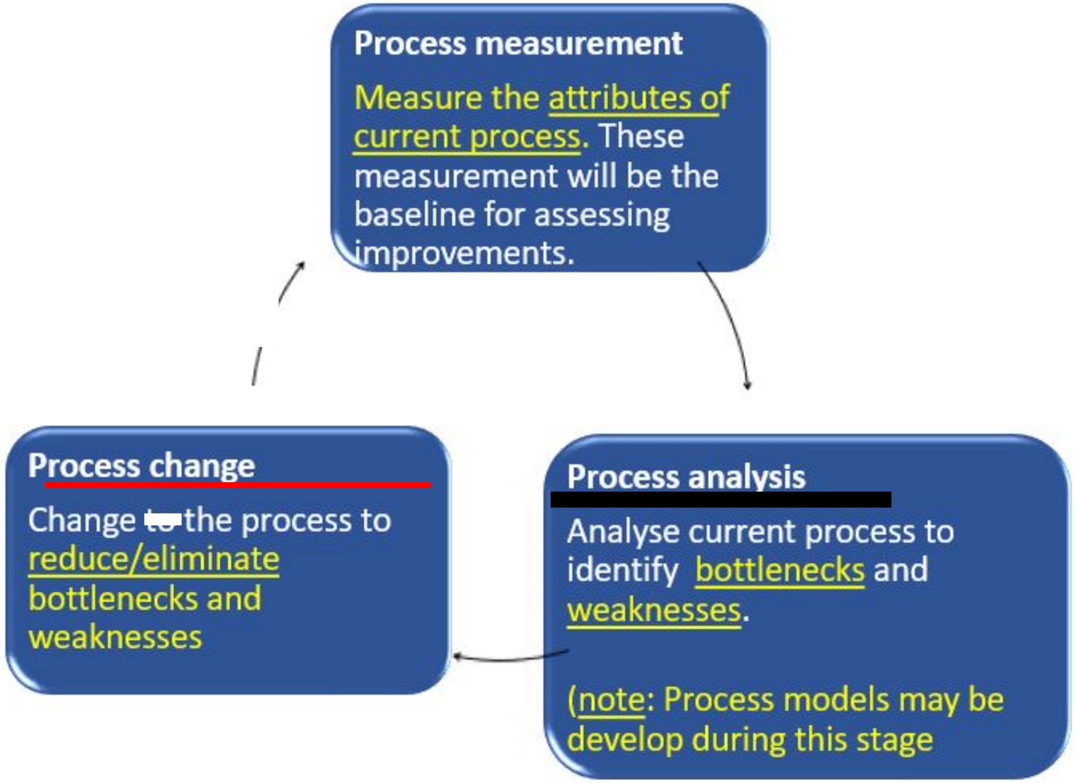
*Helps find what's **missing, inefficient, ignored**, etc in a process ______ trying to improve it.*

# Process Change

**Modify existing processes.**

may involved:

- **Introducing new practices, methods or processes;**
  *e.g. conduct peer review*
  ,
- **Change the order of process activities;**
- **Introduce new or remove existing deliverables;**
- **Introduce new roles/responsibilities.**
  *e.g. - form quality circle team (meet up weekly to solve work related problems)*
  *- write lesson learnt report*

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## Techniques

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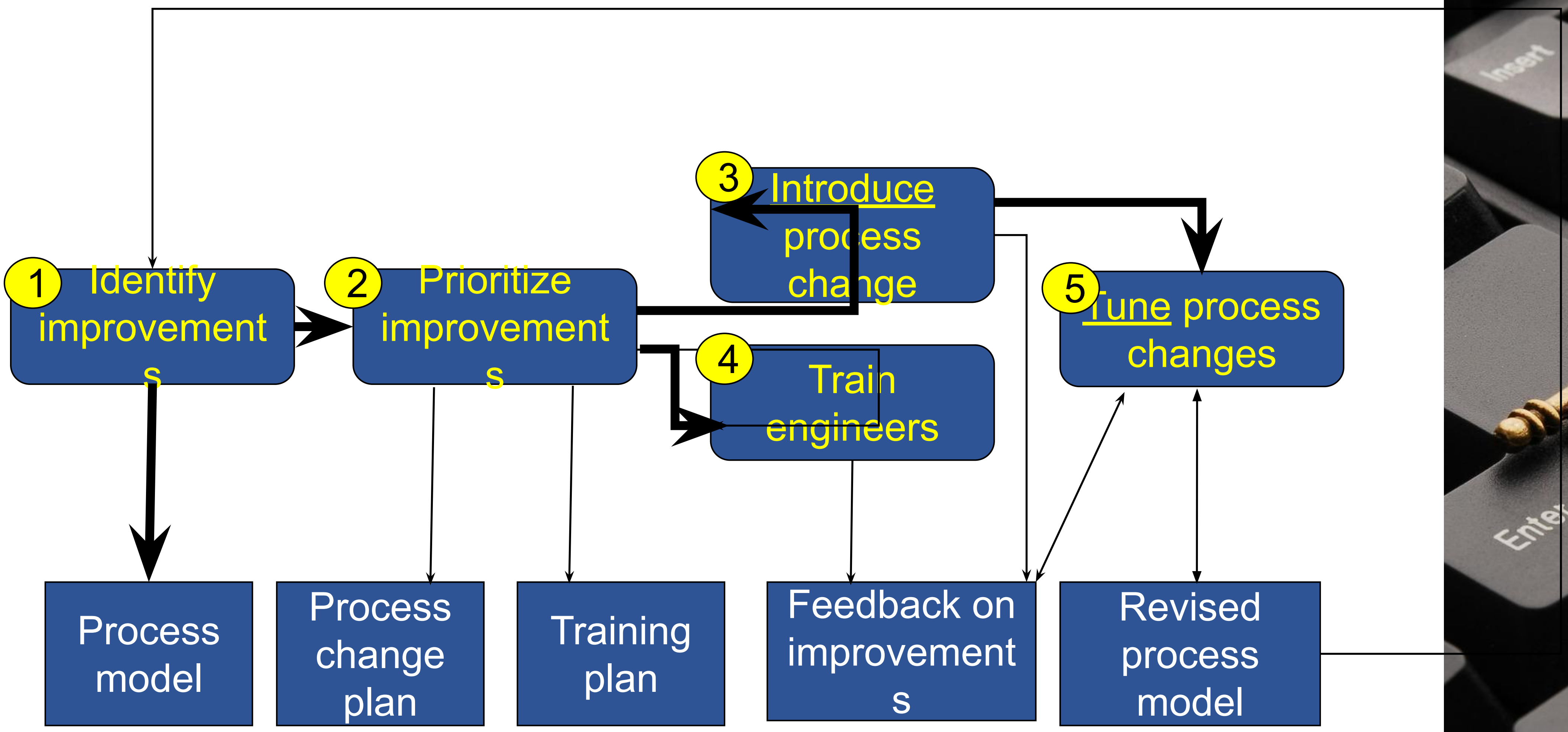
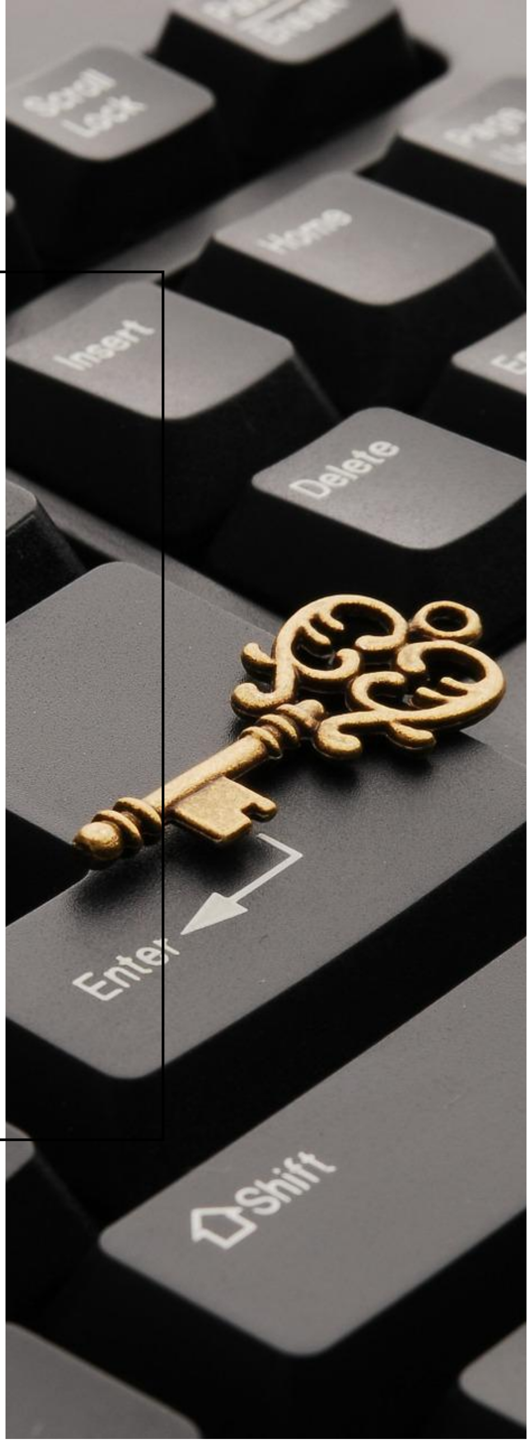
# Process Change - Process

6.2 Software Process Improvement Cycle

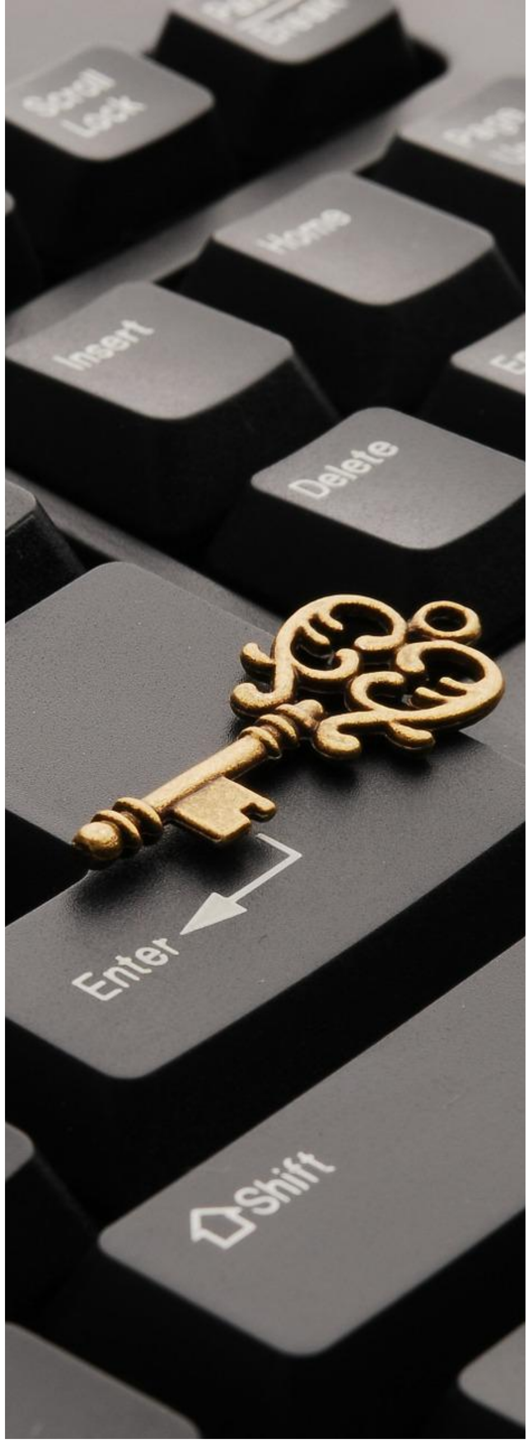


# Process Change – 5 Stages

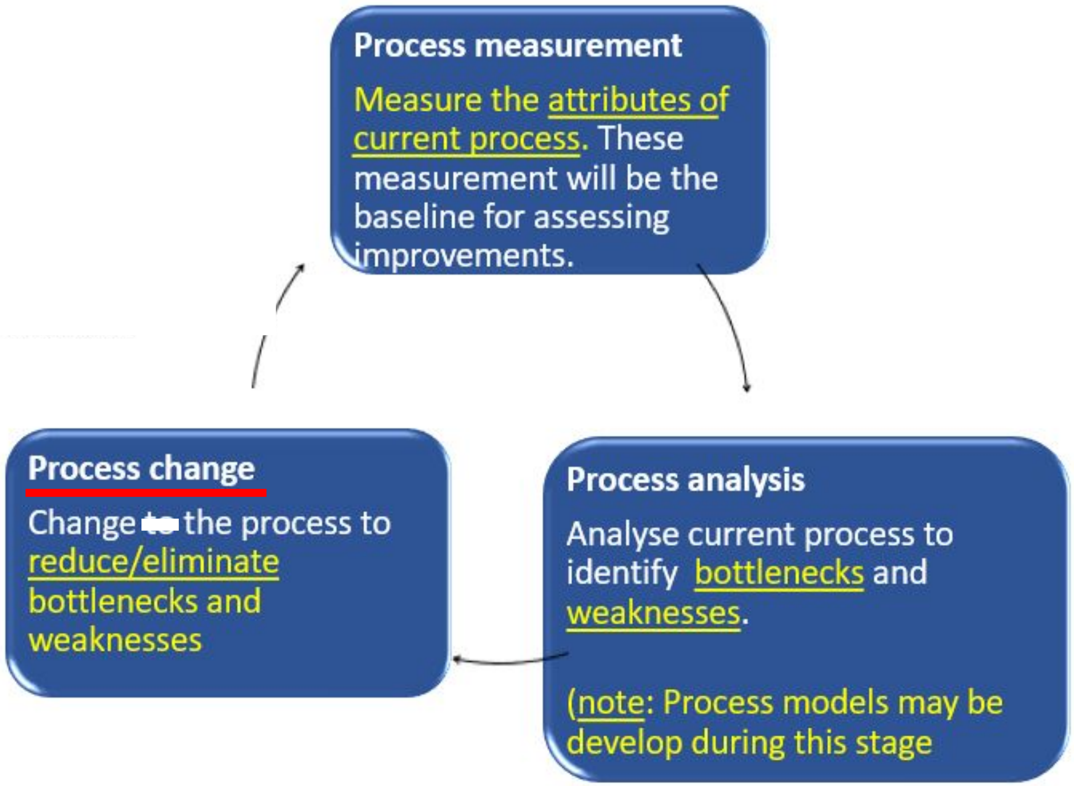
6.2 Software Process Improvement Cycle

1. **Identify Improvements**: use the results of the process analysis to identify ways to tackle quality problems, schedule bottlenecks or cost inefficiencies that have been identified during process analysis.
2. **Prioritization Improvements:** when many possible changes have been identified, it is usually impossible to introduce them all at once, and you must decide which are the most important.
3. **Introduce Process Change**: put new procedures, methods and tools into place and integrating them with other process activities.
4. **Train Engineers**: needed to gain the full benefits of process changes. The engineers involved need to understand the changes that have been proposed and how to perform the new and changed processes.
5. **Tune Process Changes:** Proposed process changes will never be completely effective as soon as they are introduced. You need a tuning phase where minor problems can be discovered, and modifications to the process can be proposed and introduced.




# Process Change – Difficulties

## 6.2 Software Process Improvement Cycle

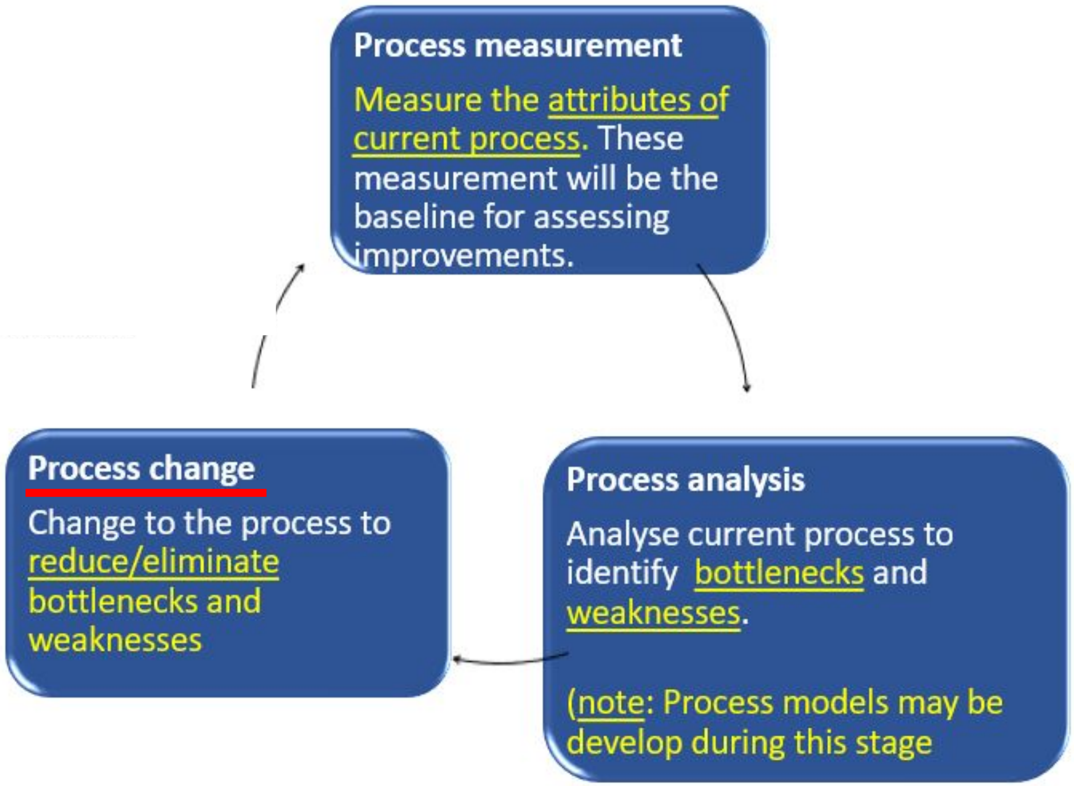



### 1. Resistance to change

- Project managers resist process change because **any innovation has unknown risks** associated with it.
  - Project managers are judged according to whether or not their project produces software on time and to budget.
  - **PM may prefer an inefficient but predictable process than an improved process** that has organizational benefits, but which has short-term risks associated with it.
- **Engineers may resist** the introduction of new processes **for similar reasons or because they see new processes as threatening their professionalism.** *(they may feel that the new process gives them less discretion and does not recognize the value of their skills and experience)*

# Process Change – Difficulties

6.2 Software Process Improvement Cycle




## 2. Change Persistence

The problem of changes being introduced then subsequently discarded

- Changes may be proposed by an '**evangelist**' who believes strongly that the changes will lead to improvement. He or she may work hard to ensure the changes are effective and the new process is accepted.
- If the 'evangelist' leaves the company, **then the people involved may simply revert to the previous ways of doing things.**

Process change must not be dependent on individuals.

Change should be officially enforced by the organization i.e. make the change permanent by making it a standard practice in the company, with company-wide support and training.

# 6.3 Process Exceptions

- Software processes are complex and process models cannot effectively represent how to handle exceptions:
  - Several key people becoming ill just before a critical review;
  - A breach of security means that all external communications are out of action for several days;
  - Organisational reorganisation;
  - A need to respond to an unanticipated request for new proposals.
- Under these circumstances, the model is suspended and managers use their initiative to deal with the exception.
- There are too many exceptions cases, so **PM needs to be flexible in handling exceptions and dynamically change the “standard” process to cope exceptional situation.**

# 6.2 Software Process Improvement Cycle - A Brief Summary

**Process measurement**

Measure the attributes of current process. These measurement will be the baseline for assessing

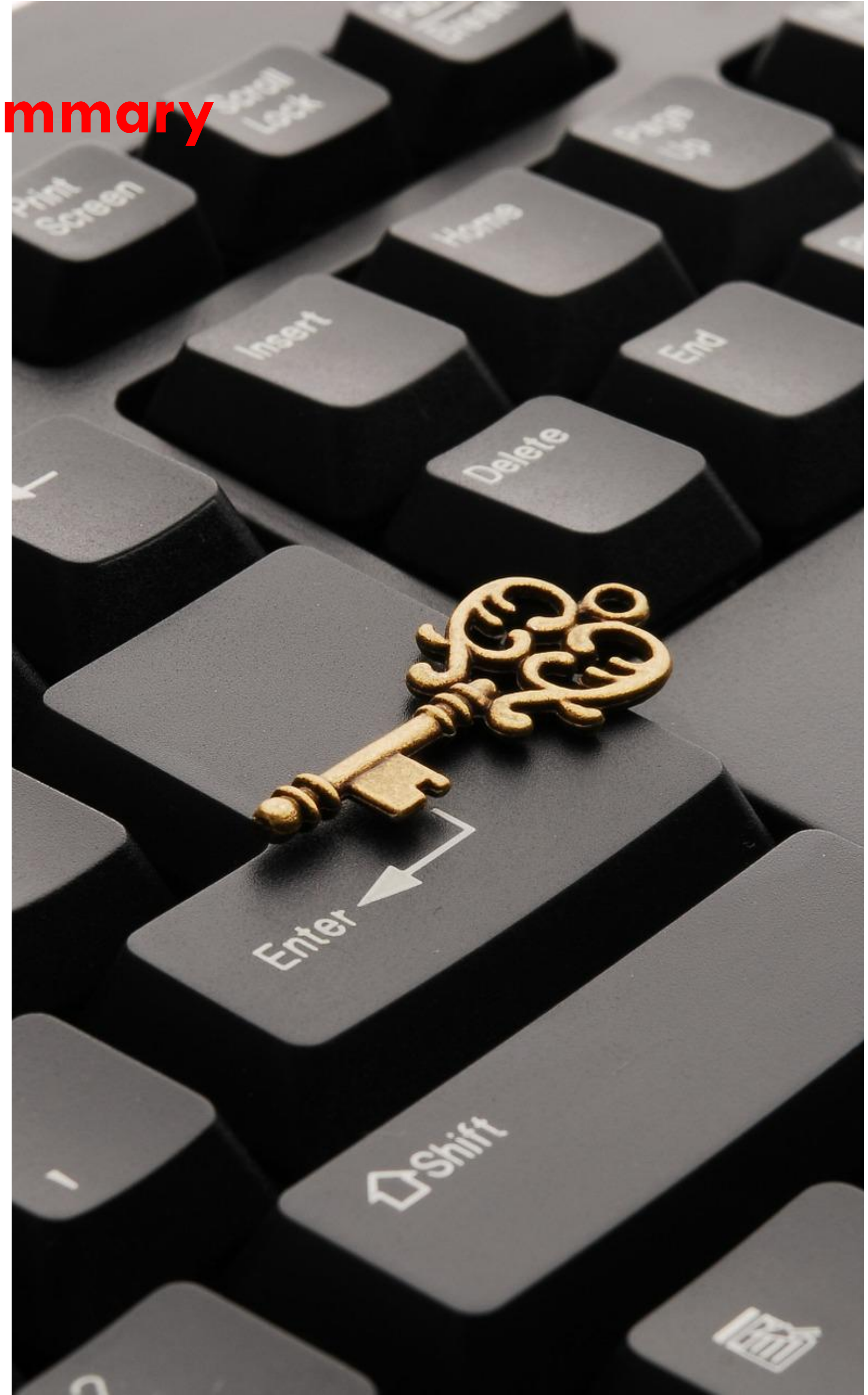
**Process analysis**

Analyse current process to identify bottlenecks and weaknesses.

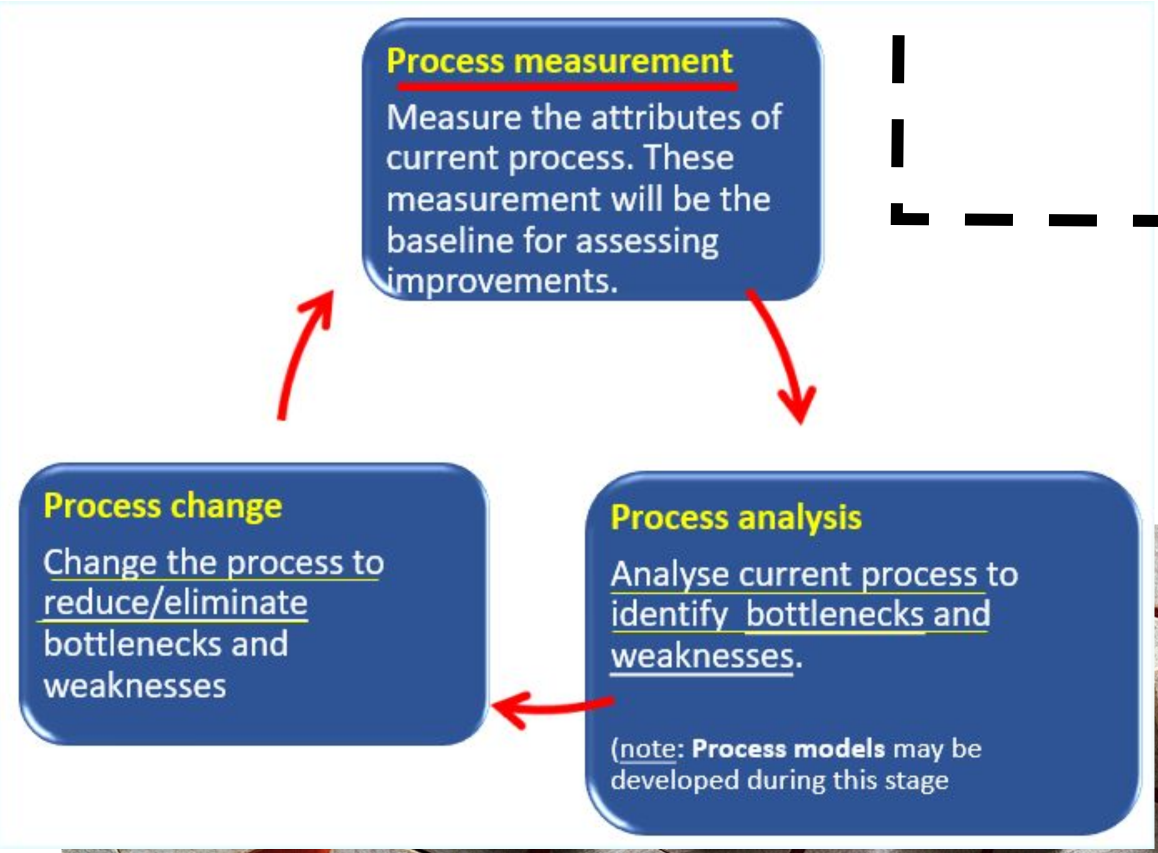
(note: **Process models** may be developed during this stage

**Process change**

Change the process to reduce/eliminate bottlenecks and weaknesses



## 6.2 Software Process Improvement Cycle




## 6.4 The CMMI Process Improvement <u>Framework</u>

*(Capability Maturity Model Integration)*

- CMMI Overview
- CMMI Process Areas
- CMMI **<u>Staged</u> Model**
- CMMI **<u>Continuous</u> Model**
- CMMI Continuous vs Staged Models

Provides guidance on improving:

- **Processes**
- **Project management**

**Also aim to introduce good software engineering practice.**

# CMMI Overview

6.4 The CMMI Process Improvement Framework

- The Capability Maturity Model Integration (CMMI) Process Improvement Framework enable integration of source models such as the CMM for Software (SW-CMM), Integrated Product Development CMM (IPD-CMM), etc. developed by Carnegie-Mellon University's Software Engineering Institute (SEI)
- An evolutionary improvement path for software organizations and includes descriptions the key elements of an effective software process.
- Provides guidance on how to gain control of processes for developing and maintaining software.
- CCMI representations include:
  - Staged Model/Representation (Maturity Levels)
  - Continuous Model/Representation (Capability Levels)


# CMMI Overview - Representation

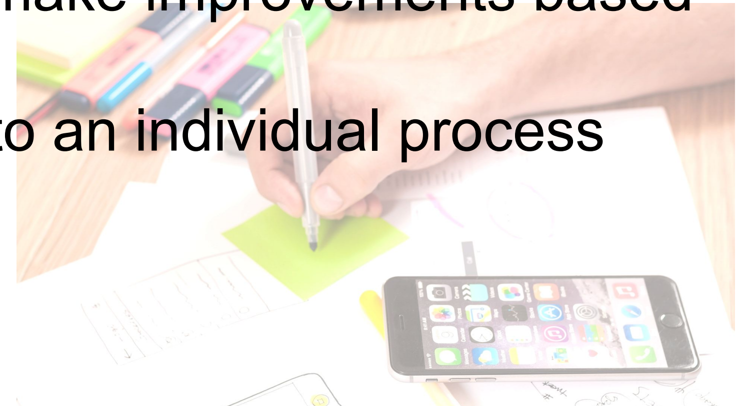
## 6.4 The CMMI Process Improvement Framework

**a.** CMMI Staged Model (Maturity Levels)

- Used in Software Capability Maturity Model (CMM)
- Uses predefined sets of process areas to define an improvement path for an organization.
- This improvement path is described by the Maturity Level model component, a well defined evolutionary plateau towards achieving organizational processes.

b. CMMI Continuous Model (Capability Levels)

- Used in Systems Engineering Capability Model (SECM) and the IPD-CMM
- Allows an organization to select a specific process area and make improvements based on it.
- Uses Capability Levels to characterize improvement relative to an individual process area.

# CMMI Process Areas

## 6.4 The CMMI Process Improvement Framework

- A Process Area is a cluster of related practices in an area that, when implemented collectively, satisfy a set of goals considered important for making significant improvement in that area.
- The continuous representation enables the organization to choose the focus of its process improvement efforts by choosing those process that best benefit the organization and its business objectives.


# CMMI Process Areas – Generic & Specific Goals



- Each process area is defined by a set of goals and practices. There are two categories of goals and practices −
  - Generic goals and practices − They are a part of every process area.
    *(Refer to the following slide)*
  - Specific goals and practices − They are specific to a given process area.
    *(Refer to Appendix 6.1)*

- A process area is satisfied when the processes of a company cover all of the generic and specific goals and practices for that process area.


# CMMI Process Areas – Generic Goals & Practices

6.4 The CMMI Process Improvement Framework

- Generic goals (GG) & generic practices (GP) are a part of every process area.

GG 1 Achieve Specific Goals
- GP 1.1 Perform Specific Practices

GG 2 Institutionalize a Managed Process
- GP 2.1 Establish an Organizational Policy
- GP 2.2 Plan the Process
- GP 2.3 Provide Resources
- GP 2.4 Assign Responsibility
- GP 2.5 Train People
- GP 2.6 Manage Configurations
- GP 2.7 Identify and Involve Relevant Stakeholders
- GP 2.8 Monitor and Control the Process
- GP 2.9 Objectively Evaluate Adherence
- GP 2.10 Review Status with Higher Level Management

GG 3 Institutionalize a Defined Process
- GP 3.1 Establish a Defined Process
- GP 3.2 Collect Improvement Information

GG 4 Institutionalize a Quantitatively Managed Process
- GP 4.1 Establish Quantitative Objectives for the Process
- GP 4.2 Stabilize Sub process Performance

GG 5 Institutionalize an Optimizing Process
- GP 5.1 Ensure Continuous Process Improvement
- GP 5.2 Correct Root Causes of Problems

# CMMI Process Areas – Common Areas

## 6.4 The CMMI Process Improvement Framework

- The common features (attributes that indicate whether the implementation and institutionalization of a key process area is effective, repeatable, and lasting) are:

Commitment to Perform: Commitment to Perform describes the actions, the organization must take to ensure that the process is established and will endure. Commitment to Perform typically involves establishing organizational policies and senior management sponsorship.

Ability to Perform: Ability to Perform describes the preconditions that must exist in the project or organization to implement the software process competently. Ability to Perform typically involves resources, organizational structures, and training.

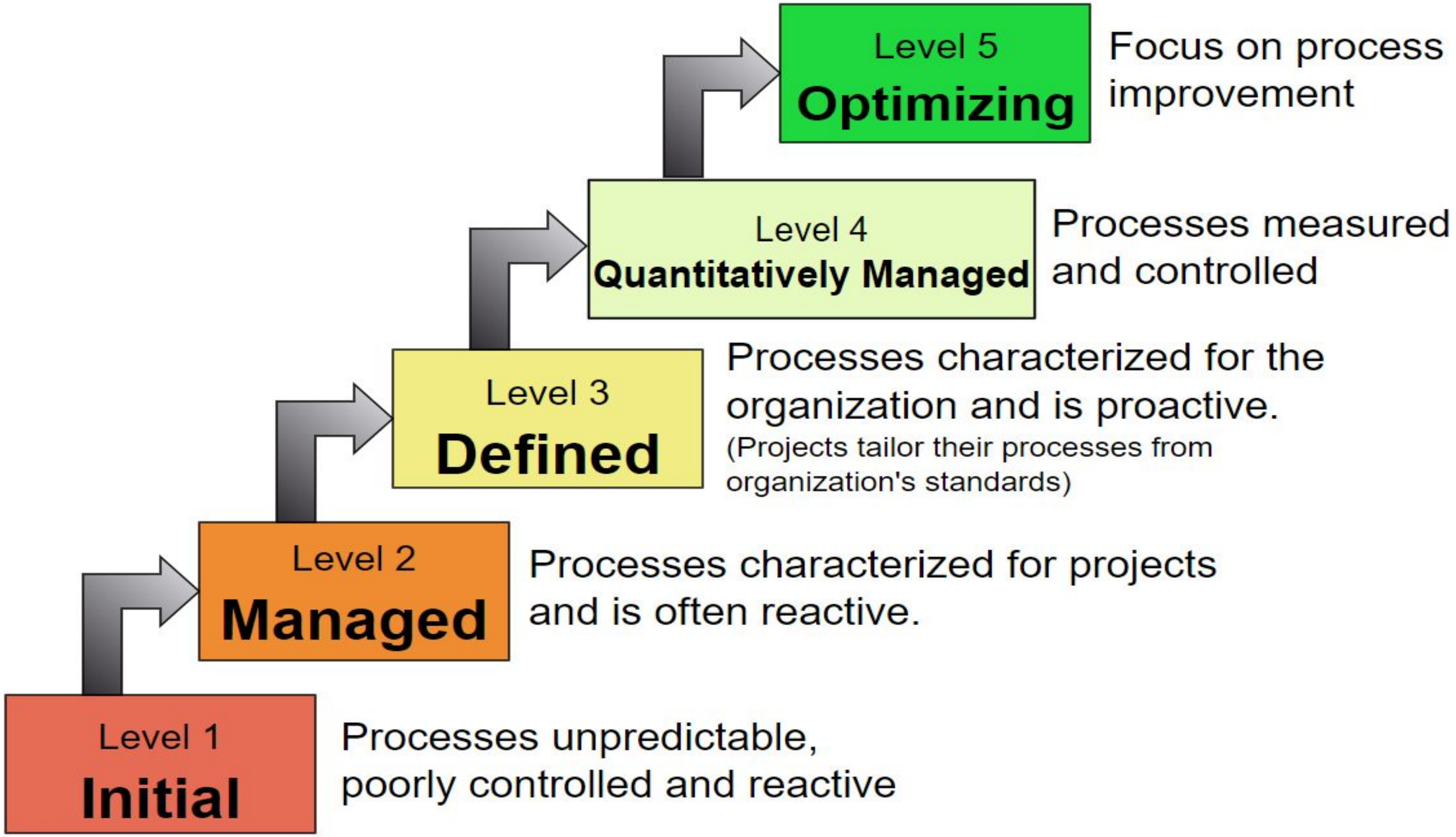
Activities Performed: Activities Performed describes the roles and procedures necessary to implement a key process area. Activities Performed typically involve establishing plans and procedures, performing the work, tracking it, and taking corrective actions as necessary.

Measurement and Analysis: Measurement and Analysis describes the need to measure the process and analyze the measurements. Measurement and Analysis typically includes examples of the measurements that could be taken to determine the status and effectiveness of the Activities Performed.

Verifying Implementation: Verifying Implementation describes the steps to ensure that the activities are performed in compliance with the process that has been established. Verification typically encompasses reviews and audits by management and software quality assurance.

# CMMI Staged Model - Maturity Levels Characteristics

6.4 The CMMI Process Improvement Framework

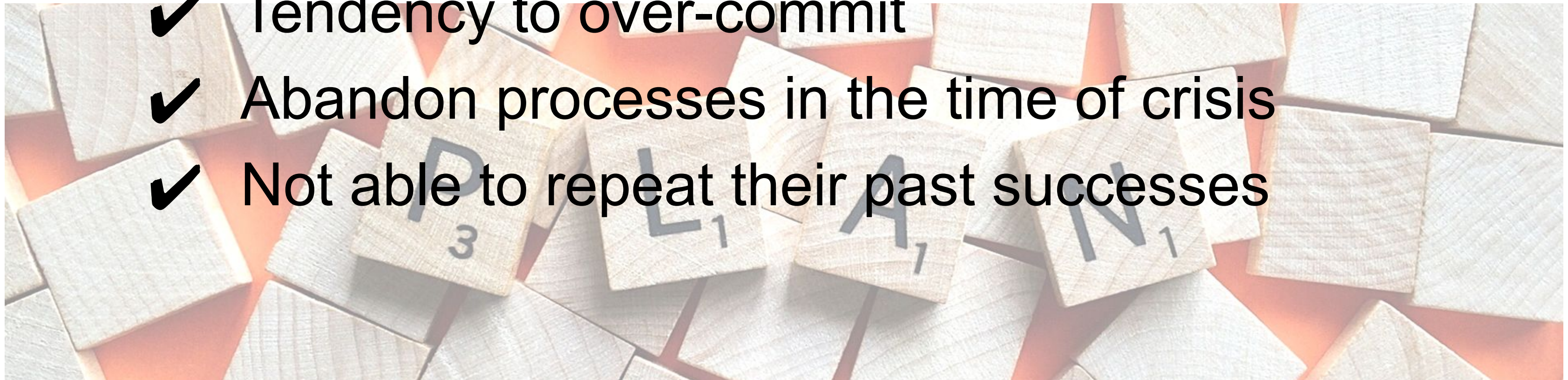



Credit: Sally Godfrey - What is CMMI ?, Public Domain, https://commons.wikimedia.org/w/index.php?curid=16790845

# CMMI Staged Model - Maturity Levels 1 Initial

6.4 The CMMI Process Improvement Framework

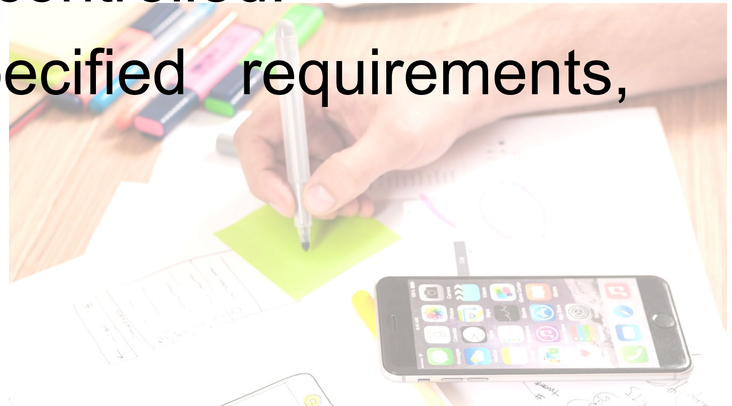
- Processes are usually ad hoc and chaotic.
- Success of the organization depends on the competence of the people in the organization and not on the use of proven processes.
- The organization usually do not provide a stable environment. Instead, software project success depends on having quality people.
- The organization often produce products and services that work. However, they frequently exceed the budget and schedule of their projects.
- Therefore, the organization is characterized by:
  - ✔ Tendency to over-commit
  - ✔ Abandon processes in the time of crisis
  - ✔ Not able to repeat their past successes



# CMMI Staged Model - Maturity Levels 2 Managed

6.4 The CMMI Process Improvement Framework

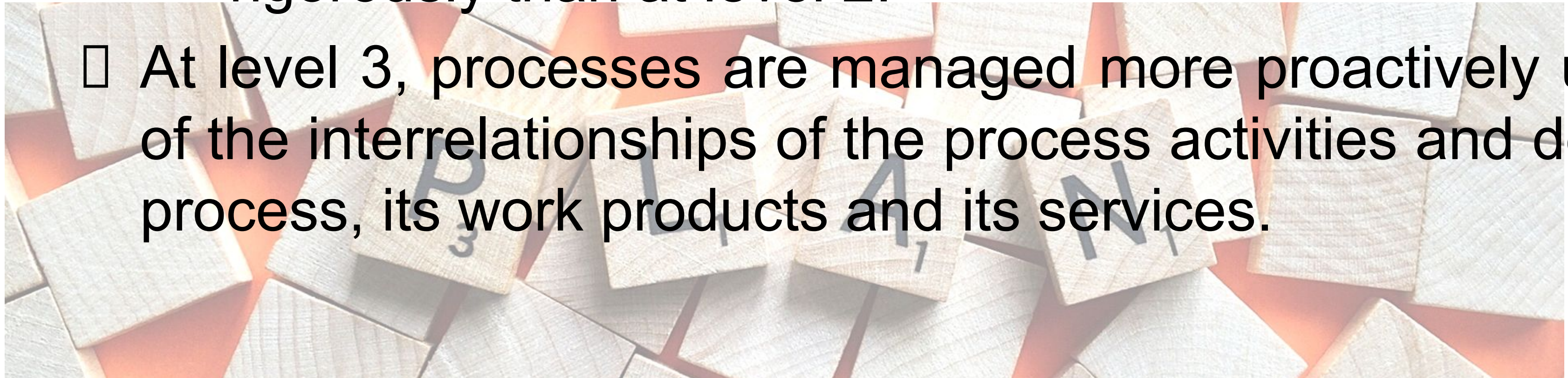
- The organization has ensured that requirements are managed and that processes are planned, performed, managed and controlled.
- The process discipline reflected in this level helps to ensure that existing practices are retained during times of stress. When these practices are in place, projects are performed and managed according to their documented plans.
- The status of the work products and the delivery of services are visible to management at defined points.
- Work products are reviewed with stakeholders and are controlled.
- The work products and services satisfy their specified requirements, standards and objectives.

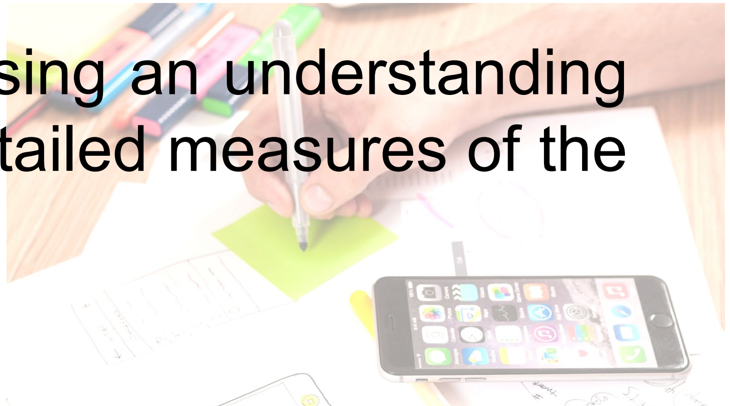


# CMMI Staged Model - Maturity Levels 3 Defined

## 6.4 The CMMI Process Improvement Framework

- Processes are well-characterized and understood, and are described in standards, procedures, tools and methods.
- Difference between level 2 and 3:
  - At level 2 the standards, process descriptions, and procedures may be quite different in each specific project. At level 3. the standards, process descriptions, and procedures for a project are tailored from the organization’s set of standard processes to suit a particular project.
  - At level 3, processes are typically described in more detail and more rigorously than at level 2.
- At level 3, processes are managed more proactively using an understanding of the interrelationships of the process activities and detailed measures of the process, its work products and its services.





# CMMI Staged Model - Maturity Levels 4 Quantitatively Managed



- Subprocesses are selected that significantly contribute to overall process performance. These selected subprocesses are controlled using statistical and other quantitative techniques.
- Quantitative objectives for quality and process performance are established and used as criteria in managing processes. Quality and process performance are understood in statistical terms and are managed throughout the life of the processes.
- Special causes of process variation are identified and, where appropriate, the sources of special causes are corrected to prevent future occurrences.
- Quality and process performance measures are incorporated into the organization's measurement repository to support fact-based decision making in the future.


# CMMI Staged Model - Maturity Levels 4 Quantitatively Managed

6.4 The CMMI Process Improvement Framework

Difference between Level 3 and 4:

- At level 4, the performance of processes is controlled using statistical and other quantitative techniques, and is quantitatively predictable. At level 3, processes are only qualitatively predictable.


# CMMI Staged Model - Maturity Levels 5 Optimizing

## 6.4 The CMMI Process Improvement Framework

- Processes are continually improved based on quantitative understanding of the common causes of variation inherent in processes.
- Focuses on continually improving process performance through both incremental and innovative technological improvements.
- Difference between level 4 and 5:
  - ✔ At level 4, processes are concerned with addressing special causes of process variation and providing statistical predictability of the results. Though processes may produce predictable results, the results may be insufficient to achieve the established objectives.
  - ✔ At level 5, processes are concerned with addressing common causes of process variation and changing the process to improve process performance in order to achieve the established quantitative process-improvement objectives.

# CMMI Staged Model – Process Areas

6.4 The CMMI Process Improvement Framework

Each CMM maturity level (ML) is accompanied by **Key Process Areas (KPA)** which must be met before

Adhoc means: Work is not planned!
(e.g. later pls attend project progress meeting at 4pm)

| Level | | | Result |
|---|---|---|---|
| 1<br>Initial | Process is informal & Adhoc | | Lowest quality / Highest risk |
| 2<br>Managed | Basic Project Management | • Requirements Management<br>• Project Planning<br>• Project Monitoring & Control<br>• Measurement & Analysis<br>• Process & Product Quality Assurance<br>• Supplier Agreement Management<br>Configuration Management | Low quality / High Risk |

Company at level 2 have develop competencies in these process areas


# CMMI Staged Model – Process Areas

6.4 The CMMI Process Improvement Framework

| Level | Focus | Key Process Area | ~~Result~~ | Result |
|---|---|---|---|---|
| 3 Defined | Process Standardization | • Requirements Development<br>• Technical Solution<br>• Product integration<br>• Verification<br>• Validation<br>• Organizational Process Focus<br>• Organizational Process Definition<br>• Organizational Training | • Integrated Project Management<br>• Risk Management<br>• Decision Analysis & Resolution<br>• Integrated Teaming<br>• Org. Environment for Integration<br>• Integrated Supplier Management | Medium quality / Medium risk |
| 4 Quantitatively Managed | Quantitatively Managed | • Organizational process performance<br>• Quantitative project management | | Higher quality / Lower risk |
| 5 Optimizing | Continuous Process Improvement | • Organizational innovation & deployment<br>• Causal analysis & resolution | | Highest quality / Lowest risk |

Company at level 3 also have competencies in these key process areas in addition to those at level 2 .

# CMMI Staged Model – Process Areas

6.4 The CMMI Process Improvement Framework

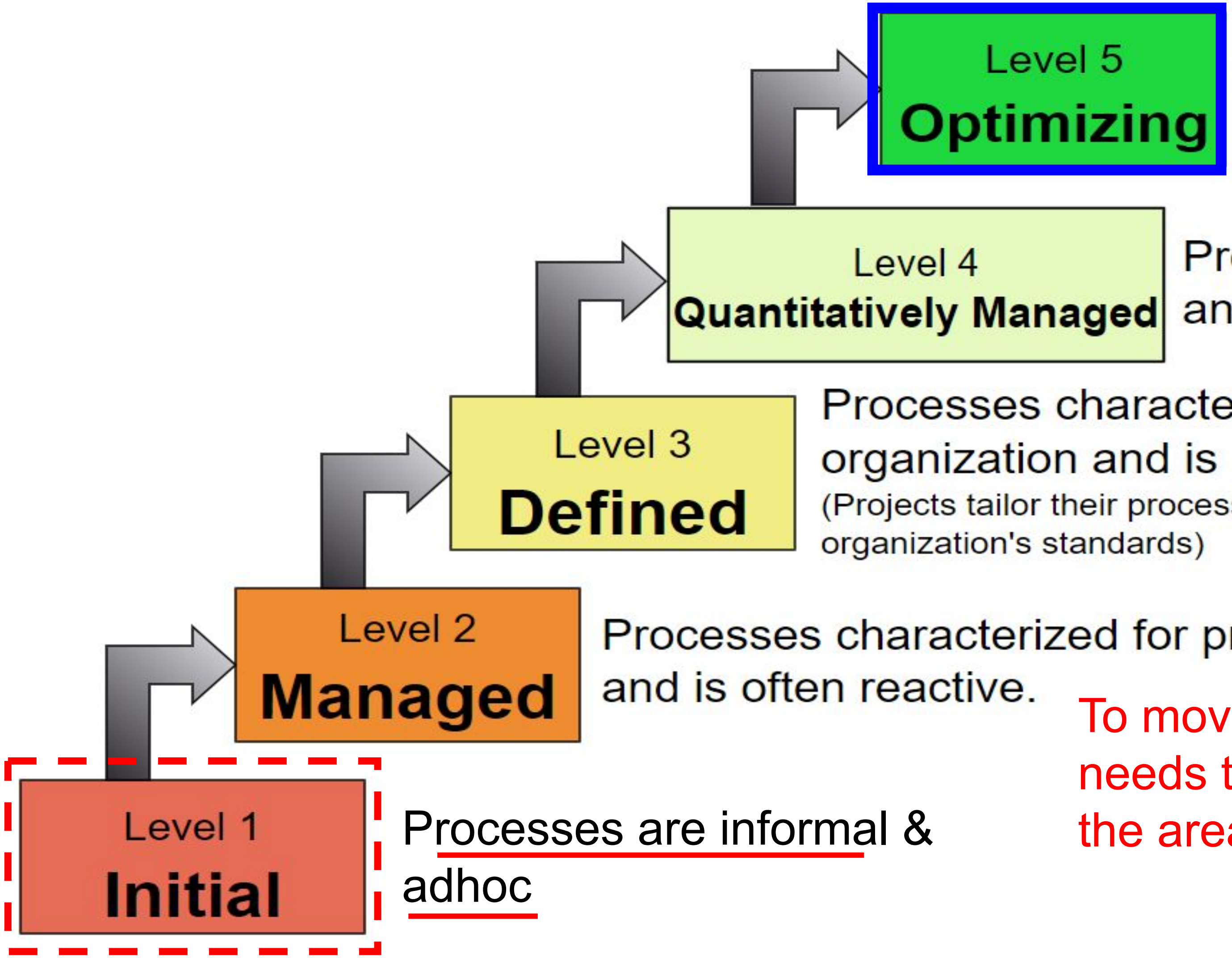
| Level | Focus | Key Process Area | Result | |
|---|---|---|---|---|
| 3 Defined | Process Standardization | • Requirements Development<br>• Technical Solution<br>• Product integration<br>• Verification<br>• Validation<br>• Organizational Process Focus<br>• Organizational Process Definition<br>• Organizational Training | • Integrated Project Management<br>• Risk Management<br>• Decision Analysis & Resolution<br>• Integrated Teaming<br>• Org. Environment for Integration<br>• Integrated Supplier Management | Medium quality / Medium risk |
| 4 Quantitatively Managed | Quantitatively Managed | • Organizational process performance<br>• Quantitative project management | | Higher quality / Lower risk |
| 5 Optimizing | Continuous Process Improvement | • Organizational innovation & deployment<br>• Causal analysis & resolution | | Highest quality / Lowest risk |

Processes are measured and controlled

# CMMI Staged Model – [Maturity Levels] Characteristics

6.4 The CMMI Process Improvement Framework

Companies at level 5 produce the highest quality of product & services. Risk is lowest.




Focus on continuous process improvement

Processes measured and controlled

Processes characterized for the organization and is proactive.
(Projects tailor their processes from organization's standards)

To move from level 2 to 3, company needs to develop competencies in the areas **specified for level 3**

Processes characterized for projects and is often reactive.

To move from level 1 to 2, company needs to develop competencies in the areas **specified for level 2**

Processes are informal & adhoc

Credit: Sally Godfrey - What is CMMI ?, Public Domain, https://commons.wikimedia.org/w/index.php?curid=16790845

# CMMI Continuous Model

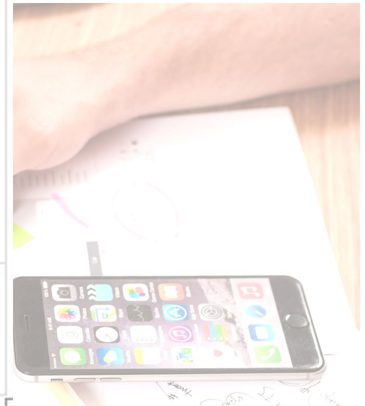
## 6.4 The CMMI Process Improvement Framework

- The continuous CMMI model does not classify an organization according to discrete levels. Rather, they are finer-grained models that consider individual or groups of practices and assess the use of good practices within each group process.
- It is a well-defined evolutionary plateau describing the organization's capability relative to a process area.
- A capability level consists of related specific and generic practices for a process area that can improve the organization's processes associated with that process area. Each level is a layer in the foundation for continuous process improvement.
  - Thus, capability levels are cumulative, i.e., a higher capability level includes the attributes of the lower levels.

# CMMI Continuous Model - Organization of Process Areas

6.4 The CMMI Process Improvement Framework

| Category | Process Area |
|---|---|
| Project Management | • Project Planning<br>• Project Monitoring and Control<br>• Supplier Agreement Management<br>• Integrated Project Management (IPPD)<br>• Integrated Supplier Management (SS)<br>• Integrated Teaming (IPPD)<br>• Risk Management Quantitative Project Management |
| Support | • Configuration Management<br>• Process and Product Quality Assurance<br>• Measurement and Analysis Causal Analysis and Resolution<br>• Decision Analysis and Resolution<br>• Organizational Environment for Integration (IPPD) |
| Engineering | • Requirements Management<br>• Requirements Development<br>• Technical Solution<br>• Product Integration<br>• Verification<br>• Validation |
| Process Management | • Organizational Process Focus<br>• Organizational Process Definition |

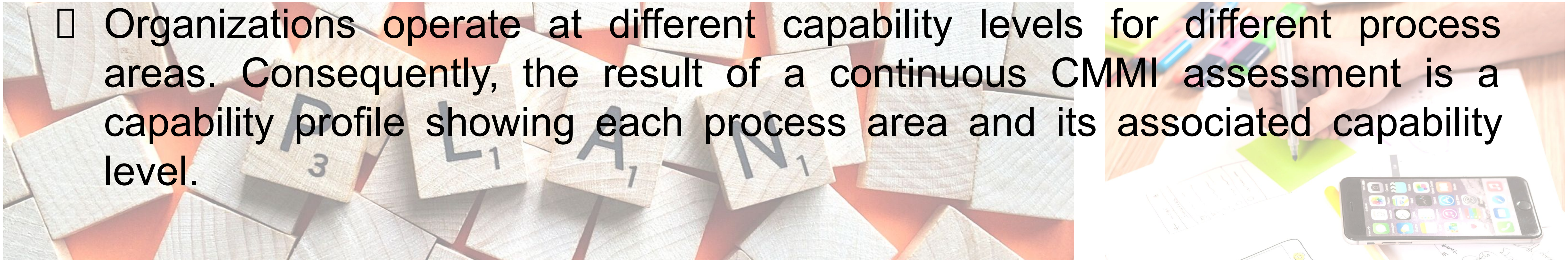


# CMMI Continuous Model – Capability Level

6.4 The CMMI Process Improvement Framework

- The continuous CMMI assigns a capability level designated by the numbers 0 through 5 to each process area.
  - 0 − Incomplete
  - 1 − Performed
  - 2 − Managed
  - 3 − Defined
  - 4 − Quantitatively Managed
  - 5 − Optimizing

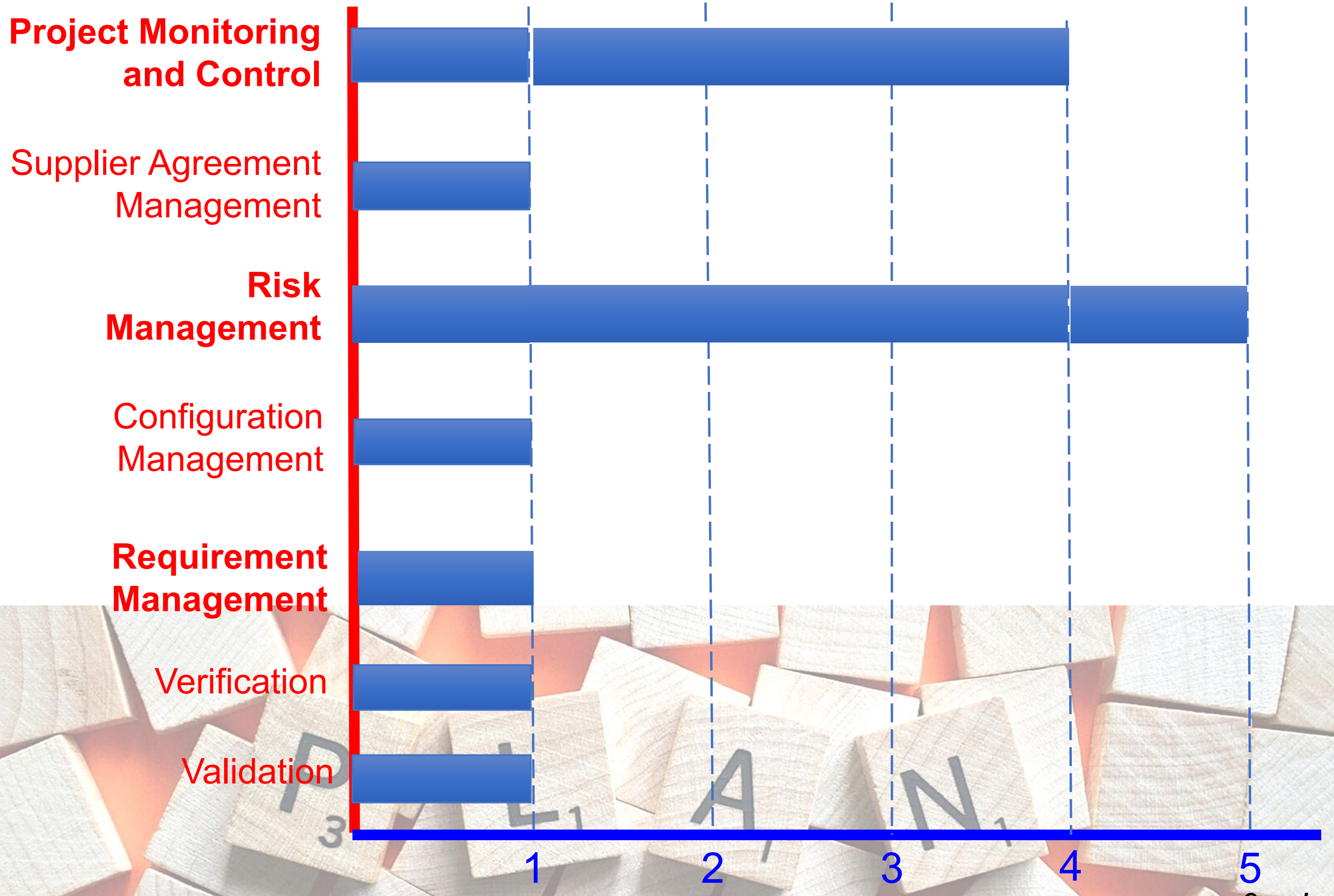
- Organizations operate at different capability levels for different process areas. Consequently, the result of a continuous CMMI assessment is a capability profile showing each process area and its associated capability level.



# CMMI [Continuous Model] – Process Capability Profile

6.4 The CMMI Process Improvement Framework

**Organizations can prioritize the key process areas they want to improve**




- In the process capability profile shown on the left, the capability level in
  - Configuration management is 4 (high)
  - Risk management is 1 (low)
- A company may develop actual and target **capability profiles** where the target profile reflects the capability level that they would like to reach for that process area.

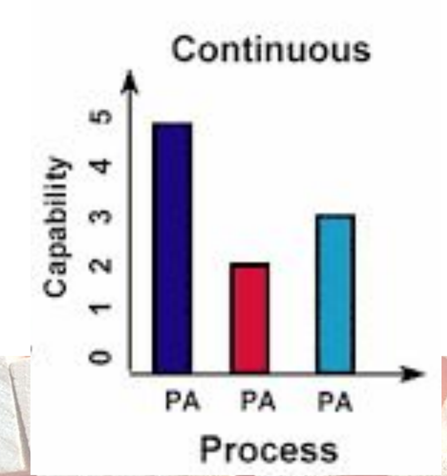
*0 – Incomplete, 1 – Performed, 2 – Managed, 3 – Defined, 4 –Quantitatively Managed, 5 – Optimizing*

# CMMI Continuous vs Staged Model

6.4 The CMMI Process Improvement Framework


## CMMI Staged Model

**Companies have <u>to follow pre-defined key process areas to develop</u>**




## CMMI Continuous Model

- Permits discretion and flexibility, while still allowing companies to work within the CMMI Improvement Framework.
- **Companies <u>can prioritise the</u> process areas for improvement according to their own needs and requirements** <u>e.g:</u>
    - A company that develops software for the aerospace industry may focus on improvements in system specification, configuration management and validation
    - A web development company may be more concerned with customer-facing processes

# Summary

## 6.1 The **Rationale for Process Improvement**

- To <u>increase</u> software **quality**
- To <u>reduce</u> development **time**
- To <u>reduce</u> development **costs**

## 6.2 Software Process Improvement **(SPI) Cycle**

*(measure □ analyse □ change)*

## 6.3 Process **Exceptions**

## 6.4 The **CMMI Process Improvement Framework**

- Stage Model
- Continuous Model

Provides guidance on improving:

- **Processes**
- **Project management**

**Also aim to introduce good software engineering practice.**